



Empirical methods in pedestrian, crowd and evacuation dynamics: Part I. Experimental methods and emerging topics

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ABSTRACT

The role of data-driven approaches in studies of crowd behaviour has received an unprecedented recognition in recent years. As a result, the size of the empirical crowd dynamics literature has rapidly grown and has more than doubled within only the last two years. This paper comprehensively reviews experimental studies of pedestrian dynamics published between April 2017 and July 2019. The aim is to capture the unprecedented growth of the experimental literature, analyse the ways in which the research landscape is changing and identify the *emerging topics*. Laboratory crowd experiments were found to be the most popular method among all empirical methods, covering the most diverse range of topics and constituting nearly half of the empirical literature. The use of animal/insect crowd experiments, on the other hand, appears to be on the decline. Virtual-reality and evacuation-drill studies have maintained their popularity. Pedestrian flow at bottlenecks, walking behaviour and route choice persist to be the top *conventional topics*. However, new research themes have emerged or have notably gained heightened momentum. This includes the evacuation of vulnerable groups (i.e. elderly, children, mobility impaired), vertical evacuation, evacuation under limited visibility, social groups and evacuation training/education. While the top conventional topics are predominantly explored through laboratory crowd experiments, the methods of evacuation drills and virtual/augmented reality are proving more instrumental in studying the top emerging topics. This highlights the complementary relation of various experimental approaches in this field. More work seems to be needed to establish the transferability of experimental findings across methods, geometric layouts, contexts and demographics.

1. Introduction

The research in the field of crowd dynamics, as an interdisciplinary and rapidly growing research domain, is intended to enhance the safety of pedestrian human by understanding the underlying mechanisms of moving crowds, the phenomena emerged from their pedestrians' collective motion and the decision-making behaviour of people within the crowd. This has created a broad array of research questions ranging from aspects related to the mechanical movement of humans in a crowd to the psychological and behavioural aspects of moving pedestrians. This diversity of research questions has led to the development of a distinctly interdisciplinary research field that engages a broad variety of expertise including, and not limited to, mathematics, computer science, traffic and transport engineering, statistical physics, sociology and psychology. It is expected that the collective knowledge emerged from these studies could ultimately lead to methodological developments, including computational tools, that would enable practitioners accurately predict crowd movements. This would help them enhance the safety as well as the physical and psychological comfort of people within crowds, prepare for instances of emergency evacuations in public crowded spaces, test the effectiveness of management policies and come up with effective measures that aim to prevent crowd crushes

or other incidents alike.

While it is hard to draw clear boarders for the academic literature of this fast-growing and multi-faceted research area, the research in this domain is often and commonly characterised by the overarching title of "pedestrian, crowd and evacuation dynamics". This encompasses any computational or empirical, qualitative or quantitative study that explores or models the behaviour of moving pedestrians in emergency or non-emergency conditions. Traditionally, this research has been perused in two major avenues: studies whose aim is to mathematically model crowd movement and those that experiment or collect field data on various aspects of crowd motion. While the literature in the past had demonstrated some degrees of divide and disconnect between these two subdomains, it appears that the gap is gradually narrowing. Great efforts are being made so that the mathematical formulations and equations of mathematicians and physicists be informed or calibrated by the data collected from empirical studies (Kretz et al., 2018; Sparnaaij et al., 2019) while also taking into account qualitative theories and descriptive findings of psychologists in model formulations (Drury, 2018; Templeton et al., 2018), before translating those equations into algorithms, computer programs, visualisation and graphics.

As a result of these developments, the empirical research in this field is gaining a notably heightened attention. It is being increasingly

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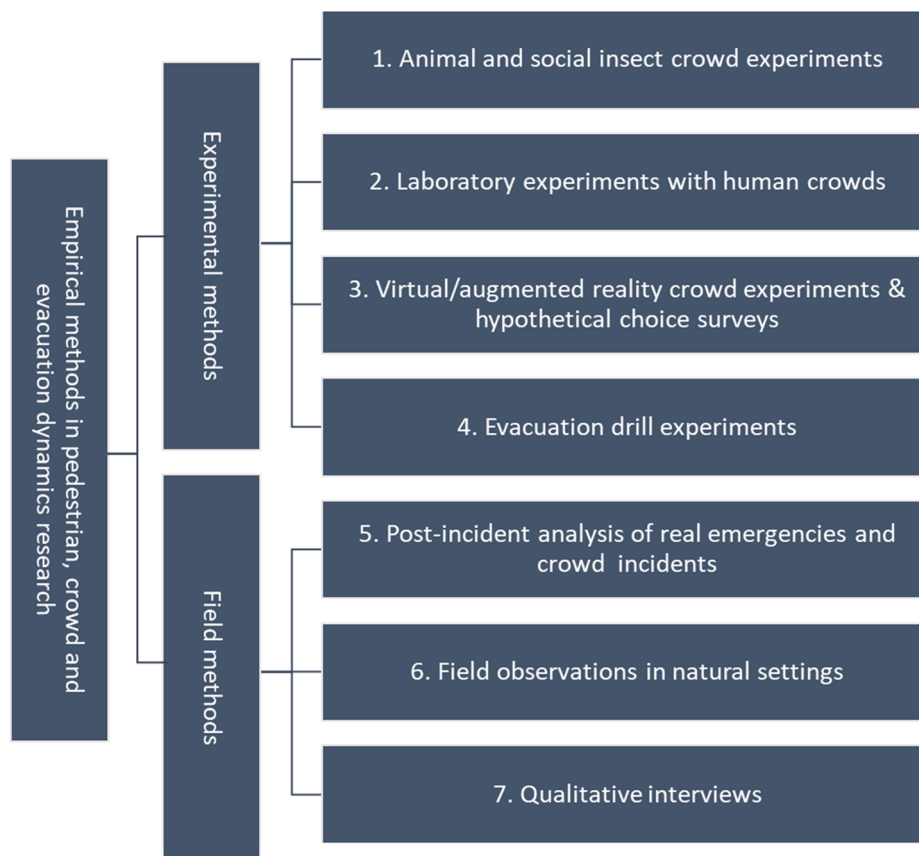


Fig. 1. Categorisation of the empirical methods in pedestrian, crowd and evacuation dynamics research.

recognised that pure arbitrary or intuition-based formulation of computational models without being rigorously tested against relevant empirical data can indeed lead to inaccurate models and misleading outcomes. It is well established among researchers of this field that this approach has the potential to lead to wrong architectural designs, ineffective policies and misleading advice to people. It is being further and further established that subtle modelling formulation or calibration flaws can indeed produce outcomes that can exert detrimental effect on crowd efficiency if applied to practice.

In addition to the fact that the quantity of empirical studies in the field of crowd research is increasing, the recent studies have shifted focus towards certain topics that did not used to receive notable attention. This appears to have created clusters of *emerging topics* that investigate phenomena beyond the simple mechanical and locomotion aspects of crowd motion, largely embodied by the *conventional topics*. It is, for example, being increasingly recognised that the assumptions that professionals and practitioners hold about possible crowd behaviour in various contexts, including emergencies, make a real difference in their policies and solutions for crowd management (Drury et al., 2019). Therefore, recent efforts in this field have more than ever brought the importance of understanding the psychological and attitudinal aspects of crowd behaviour to the attention of the researchers in this field. This seems to be generating a consensus upon the fact that the crowd dynamic research should not purely be about numerical modelling of crowd movement and locomotion aspects, and that behavioural and psychological elements could at least have equally important implications for crowd management practices. Therefore, while the recent literature is still showing growing interest in providing much-needed datasets and methods for calibration of numerical models, pure testings of behavioural theories and hypotheses are also appearing on the main research agenda due to its increasingly recognised roles in policy development (Haghani et al., 2019a). Moreover, the growing

accumulation of the experimental evidence has now generated several streams of findings that are in contradiction with previous empirical work or with previous numerical findings. This shift in the extent and landscape of the crowd dynamics research, the rapid accumulation of empirical findings, the emergence of new clusters of topics, the emergence of contradictory findings, and the changes in the relative popularity of various empirical approaches are the main motives for conducting this review study. The aim is to capture and document this rapid growth of the empirical literature and to systematically identify the changes that research themes and methods are undergoing.

Studies with empirical nature in this field were previously reviewed by Haghani and Sarvi (2018a) covering the time period of 1995 to April 2017. An analysis of the literature published since April 2017 indicates that the size of the empirical literature has more than doubled within nearly last two years. While the previous review covered nearly 140 studies, the number of studies of the same nature published since April 2017 till the time of this publication (i.e. within a time period of nearly two years), has amounted to nearly 230 new studies. It is believed that this surge in the amount of empirical work and the changes that the contemporary literature has undergone may not be properly documented or captured in the absence of a systematic and comprehensive review. Therefore, the current work could be regarded as an expansion of the previous review to the sector of the studies published since April 2017. In order to maintain consistency between these two review periods, studies are mainly categorised based on their method of data collection. The systematic analysis of the emerging and controversial topics are unique elements of the current review which have been made possible only due to the great accumulation of empirical evidence and new experimental findings in the literature. It was previously suggested that empirical methods in crowd dynamics can be differentiated in seven categories. These methods have been listed in Fig. 1. Of these seven general categories, the first four are of *experimental* nature and

the last three present more distinct characteristics of the *field* data collection. Due to the very large number of empirical studies that were published since 2017, this review is presented in two parts. Part I is dedicated to the *experimental methods* and identification of the *emerging topics*, and Part II (Haghani, 2020a) is dedicated to the *field methods* and identification of the *controversial topics* in crowd dynamics.

The following section explains the method and scope of the review and the inclusion/exclusion criteria. Section 3 provides a review of the reviews, meta-analyses and meta-syntheses on crowd dynamics published since April 2017. Sections 4–7 respectively review studies within each of the four experimental categories. While the synthesis of studies in the respective tables are presented in chronological order, they are analysed in the body of the paper in thematic clusters based on the relevance of their topics. This determines the sub-categories within each experimental method. Section 8 identifies top conventional and top emerging research themes in the contemporary crowd dynamics literature based on the surveyed studies. Section 9 summarises and concludes this review.

2. Method, scope and inclusion criteria

The search method was to large degrees consistent with that of the previous review. For the current review, however, in addition to database searching, studies were also gradually collected over time and categorised in a personal database as they appeared in the literature since April 2017. The literature was frequently monitored to identify empirical studies since April 2017. The main databases that were used for search were Google Scholar and Scopus, in addition to the personalised recommendations received by the author from sources such as ScienceDirect Message Centre or through personalised citation notifications (as a minor supplementary source for search). Overall, a variety of methods and their combination were utilised to identify the references in the new review. The time scope of the review is from April 2017 to July 2019. The search for new items stopped at the end of July 2019. In addition to that, however, during the updated search, the author came across studies published prior to April 2017 that qualified as empirical work but had been missed out in the previous review. The updated review covers these missing items from the previous review period as well so that the two reviews combined can be considered as a comprehensive review of empirical crowd research since 1995 to July 2019. Also, while the focus of the review remains on the previously mentioned time period, a selective set of articles published after July 2019, while the article was under review, are also covered in the body of the paper based on their relevance to the main topics and major research themes identified within each experimental method. The tables that synthesise the reviewed studies associated with each method and the subsequent statistical analyses remain exclusive to those published within the period of April 2017 to July 2019.

In addition to the gradual screening of the literature, in January 2019 a rather systematic search was conducted in Scopus and then was supplemented by a secondary equivalent search in Google Scholar. The search was later on updated at the end of July 2019. A general search in Scopus was first performed by using keywords combinations such as “(crowd OR pedestrian OR evacuation) AND (experiment OR empirical)”. Then subsequent search was performed using more specific keywords that would characterise each of the seven empirical methods outlined earlier individually. For category (1), for category (2), keyword combinations such as “(ant OR sheep OR mice) AND (crowd OR pedestrian OR evacuation)” was used, for category (3) combinations such as “(virtual OR augmented) AND (reality) AND (crowd OR pedestrian OR evacuation)” as well as “(hypothetical OR stated) AND (choice OR survey) AND (crowd OR pedestrian OR evacuation)” were searched. For category (4), the initial search was simply set to “evacuation AND drill” as that distinct keyword combination is commonly used and therefore can identify most of the relevant studies in this category. All keyword combination searches were extended to title,

abstract and keywords. The search items were limited to Journal articles in English (the source type). The document type only included research Article and Review. The outputs of the Scopus keyword search were individually screened and filtered by the author to check if they meet inclusions criteria. The primary inclusion criteria, apart from the time frame, was that the article had to present substantial elements of experimentation and/or data collection/utilisation. Purely numerical or purely conceptual studies that cannot fit any of the seven categories were deemed out of the scope of the review.

3. Review of reviews, meta-analyses and meta-syntheses

As it will be demonstrated quantitatively in subsequent sections in Part II of this article, the number of reviews in the field of pedestrian and crowd dynamics have substantially surged in the recent years. This is of course partly related to the rapidly growing size of the literature in this field that has made reviews and meta-analyses/syntheses necessary. These reviews have covered a broad range of topics including experimental methods (Haghani and Sarvi, 2018a; Lovreglio and Kinatder, 2020; Shi et al., 2018; Zhu et al., 2020), numerical methods (Chen et al., 2018f; Haghani, 2020b; Li et al., 2019b), choices of terminology (Adrian et al., 2019; Haghani et al., 2019b; Liu et al., 2020), qualitative theories and psychological aspects of the crowd dynamics (Drury, 2018, 2020), data challenges (Lovreglio et al., 2019; Thompson et al., 2015), and methodological challenges related to crowd research (Bode and Ronchi, 2019; Feng et al., 2018; Gwynne et al., 2017). These research efforts have made immense contributions to making the broad and cross-disciplinary literature of crowd dynamics much more organised and tractable, and as a result, more accessible to the new researchers who enter this field. The growing body of the literature has also warranted reviews on very specific and narrow topics to be published as well (Fridolf et al., 2019; Shiwakoti et al., 2019a).

3.1. Reviews on the evacuation of people with mobility limitations

Multiple reviews were identified on the topic of evacuation of vulnerable people, that is, elderly people, children and people with mobility impairment. This stream of studies has collectively called for more data-driven investigations of the evacuation behaviour of vulnerable demographic groups. Thompson et al. (2015) argue that most of the crowd movement data have been collected many years ago and are going out of date as they do not adequately represent current populations. They point out that the percentage of obese and elderly have significantly increased in the populations and that requires crowd flow datasets, models and design approaches to be re-examined, so they account for these demographic changes and the resultant reduced mobility. In a similar theme, Boyce (2017) highlight the fact that a significant portion of global population have some forms of mobility impairment whose accessibility during egress and evacuations should be taken into account. Whereas, most of our current models and estimates have been obtained from experiments with able-bodied populations, and hence not adequately reflective of the current societies. She, therefore, concludes that a revisiting of the design procedures is necessary from this perspective. It will be documented by this review how, in light of these recommendations, the contemporary studies published within the last few years have shown an unprecedented growing attention to this matter, clearly reflected in the number of publications on this topic, as one of the hottest emerging topics identified in this review (Gaire et al., 2018; Geoerg et al., 2019; Jiang et al., 2012; Kuligowski et al., 2013; Sharifi et al., 2017; Sharifi et al., 2015, 2016; Stuart et al., 2019; Tancogne-Dejean and Laclémence, 2016).

3.2. Reviews on the theories of crowd psychology

The study of Haghani et al. (2019b) revisits the terms “panic”, “irrationality” and “herding” as commonly used language in crowd

dynamics research and identifies a great degree of inconsistency in the definitions and descriptions of these terms, particularly for the first two. The authors find out that the perception of these terms and their suitability is completely different between the social and physical scientists in this field and recommended that the literature can benefit from revising these terms and using more verifiable language and definitions. As an alternative view to the panic theory, the theory of social identity was reviewed by Drury (2018). In his integrative review, he shows how an emergent sense of shared social identity can contribute to the collective coordination and self-organisation, and thereby the efficiency of crowds in mass emergencies. The study of Drury et al. (2019) overviews research on the behaviour of survivors of emergencies and disasters. They draw the key factors that explain the extent to which survivors of public emergencies cooperate with others and how this could lead to safer collective outcomes. Their review establishes that the potential role of community psychological resilience in disaster management has not been fully translated into practical guidelines. By reviewing several past studies that interviewed groups of survivors involved in emergencies, the authors provided a set of 12 recommendations for both emergency managers and members of the public affected by emergencies and disasters. The study of Ashkenazi and Hunt (2019) highlights the vital, but often overlooked, role of active bystanders (zero responders) in public emergency management practices. Their study emphasises the importance of the public, the immediate responders, in response to emergencies. They identify barriers to bystander actions and make recommendations as to how their role can be harnessed in public emergency management.

3.3. Reviews on the role of emotion and urgency in crowd evacuation response

Shipman and Majumdar (2018) focus in their review on the role of fear in humans during crowd evacuations, arguing that “evacuation under extreme situations causes specific behavioural responses”. Their review integrates the literature on this topic from multiple disciplines and covers sociological, psychological and computer modelling studies. They conclude that significant progress has been made by sociological models of human behaviour, whereas, there is a lack of appropriate software models to represent the state of fear in extreme emergencies. Following the recommendation of this review, a series of experimental work was conducted in the last two years whose aim was mainly to understand fundamental differences between evacuation behaviour under high and low states of urgency (Haghani and Sarvi, 2019a; Haghani et al., 2020; Haghani et al., 2019e). The findings of these studies will be reviewed individually in the relevant subsequent sections.

3.4. Reviews related to evacuation drill methods

The study of Gwynne et al. (2017) discusses the strengths and limitation of egress drills and their impact on egress performance and evacuation training in fire safety contexts. They explore and discuss questions such as what we know about egress drills, what their benefits are, and what the regulatory requirements of drills are in various societies. Related to the issue of drill data provision, Lovreglio et al. (2019) collect a database of pre-evacuation times from nine fire incidents and more than a hundred evacuation drills and calibrate a set of distributions for pre-evacuation time which can be used as input for numerical models.

3.5. Reviews related to virtual-reality methods

The study of Feng et al. (2018) reviews applications of immersive virtual reality serious games methods in building evacuation training and research during fire or earthquakes. It proposes a conceptual framework for effective design and implementation. Fridolf et al. (2019)

conduct a systematic review of the current experimental studies that investigated the behaviour and movement of people in smoke-filled tunnel evacuations. A large portion of these tunnel evacuation studies are based on virtual-reality experiments. Based on this review, the authors offer a design recommendation and three different methods on how to represent walking speed in both smoke-free and smoke-filled environment.

3.6. Reviews related to numerical crowd simulation methods

In a study of a relatively different theme but with same first author, Gwynne and Hunt (2018) discuss an important aspect of agent-based modelling, that is, incorporating “evacuee decision making” into the models. They outline the theoretical and practical benefits of representing the evacuee decision-making process within agent-based simulation tools. They also point out a particular benefit of including a comprehensive and explicit representation of decision-making process in agent-based models. They highlight that this allows agent-based models to serve not only as a prediction tool but also as a “computational behavioural laboratory” which can offer numerous opportunities in broadening our understanding of evacuee performance and the optimum behaviour. This means that not only can these tools be employed for prediction and planning purposes, they can be used to modify the behaviour of evacuees in computer simulated setting and observe the effect thereof on the evacuation performance, hence the name, computational behavioural laboratory. As it will be demonstrated in the subsequent sections of this article, this recommendation was followed by series of studies by the author of this current review. In those studies, a parametric agent-based model of evacuation was employed mainly as a behavioural laboratory of various aspect of evacuee decision-making (Haghani and Sarvi, 2019b,d,e; Haghani et al., 2019c). As it will be argued in those studies, these applications have great potential in helping us understand what optimum evacuation behaviour is through inexpensive computer-simulated testings. The insight from such numerical experiments can then subsequently be utilised to design actual experiments based on informed hypotheses and to eventually translate this knowledge to guidelines for efficient evacuation.

As mentioned in the introduction of this article, experimental methods have received a heightened recognition in the last few years, but a common issue relevant to many of the experimental studies is how to ‘model’ the behaviour/data observed in the experimental setting. Unless an experiment is exclusively designed for a descriptive statistical analysis, the issue of choosing the right statistical modelling approach is going to confront the analyst at some point during the study. Bode and Ronchi (2019) address this issue and provide insight and guides into model selection criteria when dealing with pedestrian movement data. Their recommendations have implications for the calibration approaches of simulation models that seek to use statistical model fitting methods. Another benchmark that is often used for calibration of numerical methods is pedestrians’ fundamental diagrams (i.e. the relationship between flow and density of pedestrians) in various walking contexts and flow configurations. The study of Vanumu et al. (2017) reviews the fundamental diagrams of pedestrian flow characteristics developed for various flow types and geometric elements.

3.7. Reviews of the force-based and cellular automata pedestrian modelling methods

Two independent reviews have addressed two prevalent numerical modelling techniques in crowd dynamics, the social force approach and the cellular automata approach. The study of Chen et al. (2018f) brings together and reviews all the modifications to the original social force model of pedestrian traffic aimed to resolve issues related to the original version and to improve its performance. They classify these improvements and assess their degree of suitability. Li et al. (2019b) similarly review cellular automata models of crowd evacuation as

Table 1
Reviews, meta-analysis, meta-synthesis studies, and discussion papers in chronological order.

Author(s) (Year)	Review/meta-synthesis?	General theme	Main focus	No. of refs	Comprehensive/selective?	Syst?	Main conclusion(s) or recommendation(s)
Thompson et al. (2015)	Review & meta-synthesis	Re-examining outdated crowd models and data	The effect of changing demographics on crowd movement and evacuation models	63	Selective	No	- Synthesising speed-flow relations for various age and body size demographics - Accounting for increasing proportion of elderly and obese people in crowd models
Wijermans et al. (2016)	Review	Real-time crowd management support systems	Developing crowd management decision-making support models	237	Selective	No	- A decision support framework, INCROWD, that guides towards more integrated support in operational crowd management - Scarcity of operational models, usable for actual crowd management
Boyce (2017)	Review	Evacuation of mixed ability populations	Design options for persons with reduced mobility	123	Selective	No	- Recognising the diverse nature of those who use our buildings and design such that everyone, irrespective of ability, can use buildings safely - Understanding of mixed ability evacuation - The need for development of computer models that can accommodate mixed ability populations
Gayathri et al. (2017)	Review	Crowd safety in mass religious gatherings	The significance of crowd management and early warning systems at mass gatherings	92	Comprehensive	No	- The importance of continuous monitoring system like an early warning system for the safety of pilgrims in mass religious gatherings
Gwynne et al. (2017)	Discussion	Egress drills	Strengths and limitations of egress drills	87	Selective	No	- Various applications of egress drill - Benefits and limitations of egress drills - Ways to fully exploit benefits of egress drills - How egress drills are used to aid research - Effectiveness of egress drills on enhancing evacuee performance - Regulation requirements of egress drills - Considerations during the design egress drills - Alternatives to egress drills for training
Vanunu et al. (2017)	Review	Fundamental diagram of pedestrian flow	Fundamental diagram for various flow types and geometric elements	58	Comprehensive	Yes	- Need for field studies of fundamental diagram - The dependency of fundamental diagram on the measurement method
Chen et al. (2018f)	Review	Social force model of pedestrian traffic	Assessment criteria for pedestrian models	75	Comprehensive	No	- Classifying the improvement and the degree of improvements for various versions of the social force model - Description ability, parameter calibration and applicability of the social force model
Drury (2018)	Review	Social identity in mass emergency behaviour	Integrating the base of the social identity research that explains collective behaviour among survivors of disasters	160	Comprehensive	No	- Explaining why people act collectively when where personal self interest is threatened - Identifying the role of social identity on the behavioural and cognitive responses - A conceptual model in which a sense of common fate is the source of an emergent shared social identity among survivors
Feng et al. (2018)	Review	Indoor building evacuations	Virtual reality serious games for evacuation training and research	68	Comprehensive	Yes	- Advantages and disadvantages of virtual reality serious games in terms of delivering evacuation knowledge - Conceptual framework to guide effective design and implementation of virtual reality methods in evacuation research and training
Gwynne and Hunt (2018)	Discussion	Agent based pedestrian simulation models	Representing evacuee decision-making process in agent-based pedestrian simulation models	51	Selective	No	- Outlining the theoretical and practical benefits of representing the evacuee decision-making - The importance of comprehensively representing evacuee decision-making in simulation models - Agent-based models can act as computational behavioural laboratory, especially when field or experimental data are hard to collect
Haghani and Sarvi (2018a)	Review	Crowd dynamics	Empirical methods in crowd dynamics	142	Comprehensive	No	- Classifying empirical methods in crowd dynamics - Identifying contradictory evidence in crowd dynamics literature - Comparing advantages and disadvantages of different empirical approaches
Shi et al. (2018)	Review	Complex movement patterns of pedestrian traffic	empirical data collection for uni- and multi-directional crowd complex movements	207	Comprehensive	No	- Challenges and opportunities on empirical studies on complex movements - Categorising complex movements of pedestrian crowd (to internal and external governed)

(continued on next page)

Table 1 (continued)

Author(s) (Year)	Review/meta-synthesis?	General theme	Main focus	No. of refs	Comprehensive/selective?	Syst?	Main conclusion(s) or recommendation(s)
Shipman and Majumdar (2018)	Review	Evacuation under extreme situations	The role of emotions (fear, stress) on human emergency response	72	Selective	No	- Empirical data rich in straight line and egress medium rich in turning, merging, weaving, and opposing flows, and poor in ingress and diverge - The disconnect between the sociological and computer models of human behaviour when it comes to representing the role of fear in human response - The need for more data to develop new theories and calibrate software models
Bode and Ronchi (2019)	Review	Statistical modelling	Statistical model fitting and model selection in pedestrian dynamics research	69	Selective	No	- Insight on choosing the most appropriate model based on data - Highlighting broad concepts, different uses, best practices, challenges and opportunities related to statistical fitting of pedestrian models
Fridolf et al. (2019)	Review and meta-analysis	Evacuation in smoke-filled underground transportation systems	Relationship between walking speed and visibility conditions	57	Comprehensive	Yes	- How people's walking speed in smoke should be represented in the fire safety design process of underground transportation systems, such as road and rail tunnels - Design recommendation on how to represent walking speed in both smoke-free and smoke-filled environments - Three different methods to represent walking speed during the design of an underground transportation system
Li et al. (2019b)	Review	Simulation of crowd evacuations	Cellular automata (CA) models of crowd evacuation	128	Comprehensive	No	- Classifying CA models into lattice gas, floor field and other field-based models - Identifying three main challenges of CA models for evacuation - Identifying advantages and disadvantages of CA models are discussed
Lovregio et al. (2019)	Review and meta-synthesis	Pre-evacuation delay	Developing a pre-evacuation database	80	Selective	Yes	- Synthesising pre-evacuation times collected from 9 fire incidents and 103 evacuation drills - Identifying sub-groups and potential factors that influence performance - Calibrating a set of pre-evacuation distributions to be used in computational models
Shiwakoti et al. (2019a)	Review	Optimizing evacuations through architectural adjustments	Effect of obstacle near exits on bottleneck flow	57	Comprehensive	No	- How and to what extent the obstacle near an exit can enhance the pedestrian outflow at the bottlenecks during emergency escape - Consensus on the beneficial effect of an obstacle - Large uncertainty on the situations on which the positive effect of obstacle could be observed - Lack of established relationship between exit width, obstacle distance and obstacle size/shape
Haghani et al. (2019b)	Review	The effect of terminology choice in crowd research	Analysing the use of the terms panic, irrationality and herding in crowd dynamics	143	Selective	No	- Lack of clear definitions for the terms panic and irrationality - The disconnect between social and physical scientists in how they perceive the panic theory - The increasing use of these terms in literature - The unsuitability of the term herding for describing the social influence phenomena
Drury et al. (2019)	Review	Social identity in public disasters	The role of survivors' cooperative behaviour in disaster response	207	Comprehensive	No	- Identifying key factors that explain why and to what extent survivors help others - Drawing recommendations for both emergency managers and public affected by emergencies and disasters to improve collective psychological resilience
Ashkenazi and Hunt (2019)	Review	The role of active bystanders in public emergencies	How active bystanders could become part of the public disaster response	13	Selective	No	- Recognising the role of bystanders as potential immediate (zero) responders in public emergencies - Identifying barriers to bystander action - How to increase likelihood of bystander intervention

another highly pervasive method in this field. They classify cellular automata models into lattice gas, floor field and other field-based models and discuss main challenges, advantages and disadvantages of each approach.

3.8. Reviews related to bottleneck flows

The study of [Shiwakoti et al. \(2019a\)](#) presents a critical review of the studies that investigated performance of an obstacle near an exit. They concluded based on this review that, while there is a relative consensus on the potential benefit of an obstacle in facilitating flow, there is still great degrees of ambiguity about the situations where a positive effect can be observable. As it will be shown in the following sections of the current review, and also in particular in the section related to the controversial topics in Part II of this article, the most recent experimental study on this topic has casted even further doubt on the previously-established assumption about the benefit of the obstacle while there are other recent studies that claim to have observed positive effects ([Garcimartín et al., 2018](#); [Shi et al., 2019](#)). This topic is among the major topics of controversy identified in this review.

A full list of the reviews and meta analyses identified in this work can be found in [Table 1](#). The table summarises the main characteristics of these studies as well as their main conclusions and recommendations.

4. Review of experiments with animal/insect crowds

Experiments with crowds of social insects and rodents, as a proxy for the movement of human crowds, continued within the period of this review, although their frequency showed a clear decline relative to the earlier period (i.e. 1995–2017). Studies that adopt this experimental technique have conventionally not shown a large degree of diversity in research questions. It is intuitively understandable that there are only a limited number of phenomena that can be experimented using this technique. These studies have in general explored three main topics: (1) phenomena related to flow at bottlenecks (i.e. faster is slower, obstacle effect, exit location effect, bottleneck shape etc), (2) symmetry breaking and (3) single-file movement, although other topics have also been occasionally investigated using this method. The new studies in this category are only limited to experiments with ants and mice, unlike the previous period in which experiments with woodlice and sheep were identified too. This review identified fourteen new studies in this category whose details are summarised in [Table 2](#). The table details various characteristics of these studies including the type of animal/insect used, the investigated phenomenon, the shape of the physical setup, the type of the repellent (the aversive stimulus) that is commonly used to get the group moving, the design factors and extracted variables. Contrary to the previous review period in which experiments of this kind were largely dominated by ant experiments, the new studies have shown more inclination to adopt experiments with rodents. Out of fourteen studies in this category, eight studies were conducted with rodents.

4.1. The flow of mouse/ant crowds through partially obstructed bottlenecks (i.e. obstacle effect)

Experiments reported by [Shiwakoti et al. \(2014\)](#), [Lin et al. \(2017\)](#) and [Wang et al. \(2018a\)](#) investigated the effect of the presence of obstacle at exit on collective egress efficiency, as a heavily-investigated question that was identified as one of the most popular topics in the previous review too ([Haghani and Sarvi, 2018a](#)). Based on ant experiments, [Shiwakoti et al. \(2014\)](#) suggested that appropriate partial obstruction of exits (i.e. with optimum size of the obstacle and distance of the obstacle to the exit) can enhance flow efficiency by a factor of two. Based on mice experiments, [Lin et al. \(2017\)](#) concluded that the presence of an obstacle in front of an exit can improve or deteriorate the evacuation efficiency depending on the physical geometry of the setup.

[Wang et al. \(2018a\)](#) performed their mice experiments under different competitiveness levels and in two different geometric setups and found out that under the low competitive condition, the obstacle has little effect on movement efficiency regardless of the geometry, but they observed an increase in the flow efficiency of nearly 10% magnitude under the competitive condition. The mixture of the evidence on this question is also in line with mixed findings of the studies that were previously reviewed on the same topic ([Haghani and Sarvi, 2018a](#)). Although the question appears to be relatively straightforward, concerning whether an obstacle can increase flow rate at exits, there appears to be many factors involved that impact on the answer. Even within the animal/insect experiments, there appears to be large degrees of inconsistency, both on the direction of the effect and its magnitude, with some studies observing only marginal or no benefit and some suggesting increased efficiency of up to 100% (as outlined above). As it will be demonstrated in detail in the section of the controversial topics in Part II of this article, the inconsistency of the findings on this question seems to even aggravate when adding the evidence from other experimental techniques, like laboratory human crowd experiments.

4.2. The faster is slower effect in ant/mouse crowd flows

Another popular topic that has been of great interest in this field, according to both previous review and the current updated version, is the effect of the competitiveness level on flow efficiency at bottlenecks, commonly referred to as the faster-is-slower effect. This topic was identified as one of the hotly debated questions in the previous review and further experimentation continued within the last two years on this topic, including those using mice methods, in particular. Experiments reported by [Oh and Park \(2017\)](#), [Zhang et al. \(2017\)](#), [Chen et al. \(2018d\)](#) all investigated this question using mice groups. [Oh and Park \(2017\)](#) recommended that agents' standard deviation of velocity can be a key factor causing faster-is-slower effect and that the effect can be eliminated by controlling the standard deviation. Clearly, the question of how standard deviation of velocities can be controlled in real practice remains to be further explored. [Zhang et al. \(2017\)](#) experimented the role of bottleneck width on the faster-is-slower phenomenon and observed that effect completely disappears for an exit wide enough to let two mice passing through side-by-side. Therefore, they recommended that the effect can be avoided by making the exit wide enough, at least twice the size of particles. This mere finding puts the practical relevance of the faster-is-slower question under question, as it makes one wonder how often do we face exits in real buildings that are narrower than the size of two average humans. The question and the recent finding related to that has very important practical implications. If an exit is wide enough to accommodate two pedestrians' side by side for example, according to this study, is it justified to advise people to not rush at exits when evacuating? Would that not have an opposite effect to what is intended, which is increasing the flow rate? [Chen et al. \(2018d\)](#) also experimented whether faster-is-slower is observable when an exit is located at the corner. Their experiment suggested that escape times do not increase by the level of stimulus when the exit is located at a corner. Therefore, they concluded that the faster-is-slower effect is not necessary a universal law for all discrete systems. As will be shown in the section related to the laboratory experiments with humans, recent empirical testings with human crowds also contributed further mixed evidence to this relatively simple but important question.

4.3. The symmetry breaking phenomenon

The studies of [Chung and Lin \(2017\)](#) and [Ji et al. \(2018b\)](#) experimented another well-known topic in crowd dynamics, the symmetry breaking. The original term and the phenomenon, as discussed in the previous review, was recommended by an experiment with groups of ants and was later on counted to be further explored using the same experimental method (see [Haghani and Sarvi \(2018a\)](#) for a review of

Table 2
Studies based on experiments with crowds of animals or social insects as a proxy for human crowds in chronological order.

Author(s) (Year)	Type of animal/insect	Method summary	Type of repellent	Physical setup	Investigated phenomenon	Primary measured variable(s)	Design factors	Main observations or applications
Shiwakoti et al. (2014)	Ants	Groups of ants escaping from single-exit chambers	Citronella	Square chamber + Circular chamber	Whether model parameters can be extrapolated from ants based on the body size ratio	- Time headways	- Presence of obstacle at exit	- A cross-species approach can provide generic insight into crowd panic
Chung and Lin (2017)	Ants	Groups of ants escaping a heated platform with two exits	Heat	Circular heated platform with two identical exits	Symmetry breaking	- Exit utilisation	- Exit location - Temperature of the platform	- The degree of asymmetry increased linearly with the temperature
Lin et al. (2017)	Mice	Groups of mice escaping a narrow exit	Smoke	Rectangular + laterally confined space	Impact of obstacle on escape efficiency	- Time intervals - Burst sizes - Evacuation time	- Presence of obstacle at exit - Distance of obstacle to exit	- Collective asymmetry is a generic phenomenon in ants - An obstacle can either reduce or increases the evacuation time depending on geometry
Oh and Park (2017)	Mice	Groups of mice escaping narrow exits connected with guide walls	Electric current	Narrow exits with guide walls	Faster is slower (FIS)	- Evacuation time - Velocity - Burst size	- Exit angle	- FIS effect observed - The increase in angle causes rodents to form lanes - Agents' standard deviation of velocity can be a key factor causing FIS - Exit angle significantly impacted on evacuation time
Shahhoseini and Sarvi (2017)	Ants	Streams of ant flows merging together	Citronella	Merging corridors	Similarity between human and ant collective motion	- Discharge rates - Time headways	- Merging angle	- Qualitative insights can be learnt about human collective movement from experiments with non-human crowds
Zhang et al. (2017)	Mice	Groups of mice escaping through narrow openings	Smoke	Rectangular chamber with one exit	Faster is slower (FIS)	- Escape times	- Exit width - Stimulus level	- FIS effect could be prevented for an exit when its width is twice the size of particles
Chen et al. (2018d)	Mice	Groups of mice escaping through narrow openings	Smoke	Rectangular chamber with one exit	Faster is slower (FIS)	- Escape times	- Exit location - Stimulus level	- Escape times do not increase obviously with the level of stimulus for the corner exit - FIS effect is not necessary a universal law
Ji et al. (2018b)	Ants	Groups of ants escaping circular vessels with two exits	None	Circular vessels with two unequal exits	Symmetry breaking	- Exit shares	- Exit widths ratio	- Asymmetric exit usage is proportional with exit widths - Similar observations across ants and human crowds
Wang et al. (2018a)	Mice	Groups of mice escaping through narrow openings	Smoke	Rectangular chamber with one exit	Impact of obstacle on escape efficiency	- Escape times	- Presence of obstacle at exit - Stress level	- Under low competitiveness, obstacle had little effect on efficiency - Under high competitiveness, obstacle reduced escape time
Wang et al. (2018d)	Ants	Groups of foraging ants moving along a narrow pipe	None	Narrow passage connecting two nests	- Bi-directional traffic - Uni-directional traffic - Conflict resolution	- Speed - Distance headway - Time to solve head-on conflicts	- None	- It took 2.61 s for unloaded ants to solve a head-on encounter - Loaded ants had a lower speed, and were less likely to be affected by head-on encounter
Wu et al. (2018)	Mice	Groups of mice escaping through narrow openings	Smoke	Rectangular chamber with one exit	Effect of exit location on evacuation efficiency	- Escape times	- Exit location	- The relocation of exit significantly impacted on the evacuation time
Zhang et al. (2018b)	Mice	Groups of mice escaping through narrow openings	Smoke	Rectangular chamber with one exit	Effect of exit width on flow dynamics	- Escape times - Flow rates - Density - Time headways	- Exit width	- Frequency of time intervals for different exit widths obeys different distributions - Total evacuation time decreased with exit width - The nonlinear relationship between width and flow

(continued on next page)

Table 2 (continued)

Author(s) (Year)	Type of animal/insect	Method summary	Type of repellent	Physical setup	Investigated phenomenon	Primary measured variable(s)	Design factors	Main observations or applications
Wang et al. (2019c)	Ants	Groups of foraging ants moving along a narrow pipe	None	Narrow passage connecting two nests	Single-file traffic	- Speed - Density - Fundamental diagram	- None	- Critical density is larger than 0.5 - The flow-density curve present as an asymmetrical arch, different to pedestrian and vehicular flow - Random pause and bodily overlap observed (different from humans) - Mice velocity increase with distance headway sharply and then keeps stable - More consistent FD between human and mice, than ants
Xiao et al. (2019a)	Mice	Groups of trained mice moving along a pipe	Water	Narrow pipe connecting two cubical rooms	Single-file traffic	- Speed - Density - Spatial headway - Fundamental diagram	- Group size	

studies on this very topic based on ant experiments published before April 2017). The two new studies mentioned above are also consistent with the same previous pattern in that they are also based on ant experiments. Chung and Lin (2017) point out to the fact that all prior experiments on this topic had adopted chemicals as the repellent. They raise the question of whether the phenomenon is also observable using heat as the stimulus for ants' movement. They observed that ants asymmetrically escaped through two exits for a certain range of temperature, and that the relation between the asymmetry level and the temperature was leaner and monotonic. Ji et al. (2018b) revisited this phenomenon noting that all prior ant experiments of symmetry breaking had been conducted in symmetric setups (i.e. two symmetry located exits with equal widths). In a series of experiments, they explored a new dimension and showed that that a class of symmetry patterns are observable for ants escaping from multiple-exit rooms while the symmetry break exists. This means that the ratio between the numbers of ants escaping from different exits is equal to the ratio between the widths of the exits. This new experimental evidence together with those on the same topic covered by the previous review make the question of symmetry breaking another controversial phenomenon in crowd dynamics.

4.4. Directional flows of ants/mouse crowds

As a new research theme using ant/mice experiments, a number of studies have reported recently on experiments of single-file and bidirectional movements, topics that used to be conventionally and almost exclusively studied in human laboratory experiments. Experiments reported in Wang et al. (2019c), Xiao et al. (2019a) and Wang et al. (2018d) are of this nature. Wang et al. (2019c) extracted fundamental diagram of the ants single-file movement based on their experiments, contrasted it with that of the humans and identified major differences in terms of the shape of the flow-density curve. In a similar context, Xiao et al. (2019a) also studied the fundamental diagram of the single-file movement but based on mice experiments. Their observation indicated that the fundamental diagram of mice is more consistent with that of humans than that of ants. Wang et al. (2018d) observed both bi-directional and uni-directional movement of ants in experimental settings and found out that their headway-speed relationship shows the same sensitivity to the distance headway in both types of flow.

4.5. Architectural designs based on ant/mouse crowd experiments

In other studies, Shahhoseini and Sarvi (2017) compared the effect of the architectural setups of the merging flows on flow efficiency across ants and humans experiments and observed consistencies across the two methods. Zhang et al. (2018b) conducted experiments with mice escaping through exits of varying widths and observed that "the frequency of time intervals obeyed a lognormal distribution or an exponential decay for different exit widths". They also observed that "the total evacuation time decreases with the increase of the exit width in a certain range. When the exit becomes wide enough, the total evacuation time increases slightly" which is in stark contrast with the body of studies that assume a linear relationship between exit width and evacuation time. Since this is one of the controversial topics that this review has identified, those studies are contrasted with one another in the section related to the controversial topics in Part II of this article.

5. Review of laboratory experiments with human crowds

Studies of laboratory experiments with human crowds were many in the period of time that is the scope of this review. More than a hundred studies of this type were identified in this survey covering a relatively broad range of topics. Certain topics, however, were of greater interest by studies that used this method, and as a result, clusters of extensively-explored topics were identifiable. Most notably, the topics of single-file

movements, step taking behaviour, fundamental diagrams, bidirectional flows, conflict avoidance and bottleneck flows were heavily studied using this method.

5.1. Single-file movement of pedestrians

Starting with the topic of single-file movement, seven experiments were reported in this review period. Jin et al. (2019a) conducted single-file pedestrian flow experiments under high-density conditions. They compared fundamental diagrams associated with the single-file movement experiments conducted in different countries. Their study is one of the few that looked into the topic of external validity and transferability of crowd experiments 'across different populations'. They also identified a critical density of about 3 ped/m. Ma et al. (2018) analysed pedestrian stepping dynamics in single-file movement and concluded that the relationship between step length (duration) and spatial headway is piecewise linear. It first increases linearly as the headway increases and then remains constant when the headway exceeds a certain amount. They also observed that the relationship between step duration (step length) and step velocity is non-monotonous (monotonous). Cao et al. (2016) experimented the stepping behaviour of pedestrian crowds with different age compositions during single-file movement. According to their findings, traffic jams occur more frequently in mixed groups due to the greater differences of mobilities and self-adaptive abilities. They also observed that the fundamental diagrams of the three age groups (young, old and mixed) are obviously different from one another. In their observation of the headway-velocity relation, three linear regimes were observed for the mixed group as opposed to only two regimes in the young student group. They furthered their analyses of this experiment in a subsequent study (Cao et al., 2018c) where the relations of velocity-step width, velocity-step length and velocity-stepping time for different age groups are discussed along with the effects of pedestrian gender and height on stepping laws and fundamental diagrams.

More recently, Cao et al. (2019c) conducted experiments to investigate dynamics of pedestrian movement in single-file movement under limited visibility and found out that pedestrians are more sensitive to the density change under good visibility. They observed typical behaviour of following the boundary and frontal pedestrian under poor visibility. Huang et al. (2018b) experimented individual and single-file pedestrian movement in narrow seat aisle and analysed interferences of inactive pedestrian from the perspective of velocity and density change. In the study of Sun et al. (2018), single-file experiments were conducted in a ship corridor for both female and male groups. They showed that density is not the only factor that affects pedestrian speed under ship trim or heeling conditions. Both individual and group walking speed were significantly attenuated by the trim angle. In another study, Huang et al. (2019) reported on experiments of one-dimensional movement of pedestrians carrying luggage. The recent experiments of Ren et al. (2019b) compared the single-file pedestrian movement of the elderly and other age groups. Their findings indicated that the age is not the only factor influencing pedestrian dynamics, rather, the composition of heterogeneous crowds and familiarity among neighbouring pedestrians also have significant effects.

5.2. Stepping behaviour of pedestrians

Further on the topic of stepping behaviour, the study of Jelić et al. (2012) investigated stepping dynamics in one-dimensional pedestrian flows. They observed that local density has a direct influence on the stepping characteristics and that the step length is proportional to the velocity of the pedestrian and is directly related to the space available in front of them. Wang et al. (2018c) and Wang et al. (2018b) reported on experimental analyses of stepping behaviour. They investigated the single-file movement of pedestrians at different densities and measured step length, step frequency, swaying amplitude and step

synchronisation using trajectories of the head. They also identified six different stepping styles that change with the density. Zeng et al. (2018) investigated frequency distribution of step length and step frequency as well the relationship between the step length and step frequency and established its dependency on the headway.

5.3. Unidirectional flows of pedestrians

Unidirectional movement of pedestrians was also even further investigated from other perspectives in addition to the studies discussed above. Jin et al. (2018) reported on the experiments of unidirectional movement at extreme density conditions (as high as 9 pedestrians per squared meters) and recorded various microscopic events. They suggested that the flow dynamics in hyper-congested state and in over-congested state are qualitatively different, and that the critical density is about 8 ped/(m²). Further analyses of these high-density experiments, including those related to bidirectional flows, can be found in the more recent study of Jin et al. (2019b). Guo et al. (2016) reported on experiments and modelling of uni-directional and bi-directional pedestrian flow in the view-limited condition. They observed quick lane formation under the normal view, but no spontaneous lane formation emerged in first set of view-limited experiment. Rather, lane formation occurred in later sets of view-limited experiments. In other studies of this theme, Zhang et al. (2018a) experimentally validated an optimization-based overtaking model for unidirectional pedestrian flow and Zhao and Zhang (2017) proposed a unified follow-the-leader social force model for vehicle, bicycle and pedestrian traffic. They proposed that the fundamental diagrams of car, bicycle and pedestrian traffic can be unified through a single scaled speed-spacing equation. More recently, Gao et al. (2019) considered the impact of carrying trolley cases on unidirectional pedestrian flows and calculated fundamental diagrams with and without trollies. Their experiments, under different combinations of corridor width and trolley-carrier ratios, indicated that in a higher density corridor, the individual velocity can be significantly affected by the trollies.

5.4. Directional flows under the rhythm effect

Studies on directional flows also presented a new interesting dimension and that was the effect of the music and rhythm on flow characteristics. In a pioneering work in this sub-domain, Yanagisawa et al. (2012) explored the possibility of improving pedestrian flow through slow rhythm. They discovered that a rhythm that is slower than normal-walking pace can increase flow if the flow-density diagram is convex downward in the high-density regime. Following that study, a more recent publication also studied the effect of background music on pedestrian movement at high density (Zeng et al., 2019). They compared fundamental diagrams with and without rhythm. These analyses suggested that velocity and flow with background music at medium and high densities will be smaller than those under normal conditions, and that, with music, stop-and-go behaviour is more frequent.

5.5. Multi-directional flows of pedestrians

The circular antipode experiments of Hu et al. (2019) focused also on the movement strategies of individuals in multidirectional (antipode) flows and identified some typical pedestrian behaviours under multidirectional flows. In their observations, detour strategy was the most efficient strategy to reach destination quickly. Conflicting flows have also been experimented in interesting setups. Bode et al. (2019) explored the emergence of macroscopic interactions between intersecting pedestrian streams (in antipode experiments) and found out that as arrival rates increase, streams start to interact and compete for space and that these interactions result in an even balance of pedestrian numbers across two orthogonally intersecting streams without either of the two streams being able to dominate the other. However, for four-

intersecting streams, they observed that jams in some streams can coincide with higher flow rates in the other streams. This way, the relative dominance of streams can switch stochastically. In another study, [Cao et al. \(2018b\)](#) compare the fundamental diagrams of Chinese and German experiments, which is another case in the list of very few studies that have thus far addressed the issue of the external validity/transferability in crowd experiments.

5.6. Conflict/collision avoidance in bidirectional pedestrian flows

[Feliciani and Nishinari \(2016\)](#) conducted an empirical analysis of the lane formation process in bidirectional pedestrian flows while changing the flow ratio across the two directions. They observed that balanced bidirectional flow is the most stable configuration. [Parisi et al. \(2016\)](#) also experimented characteristics of collision avoidance in pedestrian dynamics and avoidance behaviour under different flow rates and group sizes and during crossing and head-on conflicts. They identified two typical avoidance maneuverers, stopping and steering behaviour, and showed that the latter is more common. [Xue et al. \(2017\)](#) performed pedestrian counter flow in discrete space and time and discussed the implications for cellular automata modelling. [Yamamoto et al. \(2019\)](#) investigated the deviation between body orientation and walking direction during collision avoidance and observed that pedestrians rotated their bodies when the corridor width was smaller than the sum of the widths of the two pedestrians. More recently, the study of [Ye et al. \(2019\)](#) experimented bidirectional flows of pedestrians at right-angled corners and calculated fundamental diagrams for this type of flow.

5.7. Conflict/collision avoidance in multi-directional pedestrian flows

A considerable number of laboratory experiments investigated conflicting flows and the conflict avoidance of pedestrians. In doing so, a number of studies have been reported to have conducted their experiments in an antipode circular layout. The first of this group of publications is the study of [Wolinski et al. \(2014\)](#) who used their experimental data mainly to calibrate parameters of their conflict resolution model. The study of [Xiao et al. \(2019b\)](#) is of similar nature where the authors analysed their antipode experiment from the perspective of macroscopic indexes such as route length, route potential, travel time, velocity and time-series.

Two studies can be categorised under the general title of direction changing during walking. [Liu et al. \(2017\)](#) investigated relaxation time in direction changing movement and found that pedestrians are insensitive to tiny angular difference in freely walking. [Jia et al. \(2019\)](#) studied the evading behaviour of individual pedestrians when confronting with an obstacle in a corridor. They particularly looked into the impact of obstacle size. They observed that with the increase of obstacle width, the frequency and amplitude of body sway were barely affected, but the lateral deviation of walking direction largely increased. As this work was under review, more studies emerged on the issue of evading obstacles showing that this issue is another topic of interest that is getting traction in crowd dynamics ([Alhawsawi et al., 2020](#); [Jia et al., 2020](#)).

5.8. Merging flows of pedestrians

Similar to the previous review, a number of pedestrian merging experiments were reported in the literature. According to the experiments of [Lian et al. \(2017\)](#), lane formation was found downstream in the main channel and the increasing amplitude of the lane width formed by the branch flow became weaker with the increasing width of the branch channel. They also observed that when the branch channel was 2.4 m, the main and minor branch channel became mutual bottlenecks to each other, and that density in the corner increased when the branch width increased. [Shahhoseini et al. \(2017\)](#) compared

symmetric and asymmetric merging layouts and observed that the delay is considerably greater in asymmetrical setups compared with the delay in symmetrical setups and that the difference is even more pronounced at elevated levels of pedestrians' desired speed. [Shahhoseini and Sarvi \(2018\)](#) used micro and macro observations of the merging experiments and contrasted them with those of a numerical simulation model while observing good qualitative agreements between the two. Using an extended and more comprehensive set of experiments that examined a wide range of physical merging layouts, [Shahhoseini and Sarvi \(2019\)](#) suggested that traffic efficiency is demonstrably sensitive to the physical features of the merging passages as well as the desired speed of the crowd. The patterns of influence of the geometric constraints on traffic efficiency were by and large common across all three speed regimes.

5.9. Pedestrian flows through bottlenecks

Another heavily investigated topic using laboratory crowd experiments is the conventional bottleneck flow problem and its affiliates (i.e. the faster-is-slower effect and the obstacle effect ([Ding et al., 2020](#))). [Hoogendoorn and Daamen \(2005\)](#) showed that pedestrians inside bottlenecks effectively form layers or trails. They showed that these layers are overlapping and labelled this phenomenon as the "zipper" effect. They suggested that the zipper effect causes the capacity of the bottleneck to increase in a stepwise fashion with the width of the bottleneck. Further on the topic of capacity, [Cepolina \(2009\)](#) identify the phenomenon of "capacity drop" when high-density conditions occur upstream of a bottleneck. They propose a model that can reproduce the capacity drop phenomenon. [Liao et al. \(2016\)](#) measured the steady state of pedestrian flow in bottleneck experiments. [Bukáček et al. \(2018\)](#) conducted microscopic travel-time analyses of bottleneck experiments and suggested that the ability to push through the crowd is a combination of aggressiveness in conflicts and willingness to overtake the crowd. [Adrian et al. \(2018\)](#) performed experiments of crowding and queuing in entrance scenarios and investigated the effect of the corridor width in front of a bottleneck on the density. They showed that, in general, the density in front of the gate as well as the area of high density is increased by widening the corridor and by intensifying the motivation of the participants. Related to this topic, [Sieben et al. \(2017\)](#) analysed an experiment in which a large group of people enters a concert hall through two different spatial barrier structures. They investigated the influence of the spatial structure on the dynamics of the entrance procedure and demonstrated that density, waiting time and speed of progress show large variations. For further experiments on the inflow process of pedestrians, readers are referred to the studies of [Ezaki et al. \(2016\)](#) and [Lian et al. \(2018b\)](#). More recently, [Li et al. \(2020a\)](#) analysed bottleneck flow of preschool children in a set of laboratory experiments. They observed that the relationship between the flow and bottleneck width are similar across children and adults. A version of the abovementioned experiments with elderly pedestrians can also be found in ([Ren et al., 2019a](#)). Contrary to a body of previous studies, [Ren et al. \(2019a\)](#) observed a linear relationship between flow rate and exit width ([Gwynne et al., 2009](#)).

5.10. Pedestrian flow at bottlenecks and architectural adjustments

The study of [Tavana and Aghabayk \(2019\)](#) was also about architectural adjustments of the bottleneck and efficient design. They studied how funnel-shaped bottlenecks with different angles affect the properties of pedestrian egress flow. They suggested that the best-performing bottleneck angle follows a certain trend, with 26.6° being the most efficient angle. Further investigations on the topic of efficient design focused on the role of obstacle in front of the bottleneck and its influence on flow efficiency. [Jiang et al. \(2014\)](#) found out, based on numerical analyses, that appropriately placing two pillars on both sides (but not in front) of the door can maximize the escape efficiency and verified their findings using experiments with a group of 80

pedestrians. Shi et al. (2019) experimentally inferred that obstacles have different performance at middle and corner exits and that column size and distance to the exit affect the outflow of pedestrians. The most recent experimental evidence on the role of obstacle on flow efficiency has emerged from the studies of Zuriguel et al. (2020) and Garcimartín et al. (2018) suggesting that obstacle alleviates the pressure at the door, but, the total evacuation time, is not altered by the obstacle. Overall, their findings challenged the idea that the placement of obstacles in front of doors is an effective strategy to increase the flow of pedestrians. The most recent experiments of obstacle effect reported by Zhao et al. (2020b) has produced mixed evidence about the effectiveness of the obstacle. Their observations suggested that flow efficiency is quite sensitive to the geometrical parameters of obstacles. They also suggested that panel and double pillar are more effective than single-pillar obstacles. The recent bottleneck experiments of Jianyu et al. (2019) that investigated the effect of exit location showed that for narrow doors, the capacity decreased from the middle exit to the corner exit. But this gap gradually disappeared, as the door became wider. This constitutes another controversial topic of crowd dynamic considering the body of earlier studies that have produced evidence to the contrary (Shiwakoti et al., 2011). Also, further on the theme of revisiting previously established assumptions about the bottleneck flow, studies of Shahhoseini et al. (2018) and Haghani et al. (2019d) provided a new angle and showed that higher motivation of pedestrians to go through a bottleneck will result in faster flows, according to all measures of flow efficiency, i.e. a faster-is-faster effect, so long as the conduct of the pedestrians at the bottleneck does not amount to aggressive and explicit shoving. Haghani et al. (2019d) suggested this as one of the key determinants of faster being faster or slower at bottlenecks in addition to the observations of Garcimartín et al. (2016) that also suggested based on a temporal analysis of the flow that the number of people performing the bottleneck egress task plays a crucial role in the results.

5.11. Fundamental diagram of pedestrian flows

Studies on fundamental diagrams continued as before to be a common theme. Burghardt et al. (2013) looked at the fundamental diagrams particularly in relation to the performance of stairs. They made comparisons between the field and laboratory measurements and found out that laboratory experiments lead to slightly higher flow values than field studies. Wang et al. (2019b) discuss fundamental diagrams for different flows under all states and the steady state. They considered unidirectional flows, bidirectional counter flows and bidirectional cross flows, and four directional cross flows and analysed them using improved Voronoi diagram measurements. Ren et al. (2019c) conducted experiments of pedestrian flows in straight corridors with older adults. Through this experiment they compared fundamental diagrams of younger and older adults and observed similarities across the two age groups in terms of the shape of the fundamental diagram. Compared to young adults, the elderly showed lower free flow speed, smaller steps, lower pace and passive wait.

5.12. Pedestrian boarding and alighting behaviour

A number of studies experimented characteristics of boarding and alighting in physical models of trains or planes. The experimental study of Seriani et al. (2019) using a mock-up model of a train carriage explored the effect of boarding and alighting ratio on passengers' behaviour at metro stations. Similarly, Seriani et al. (2017) explored the pedestrian level of interaction on platform conflict areas at metro stations. Gwynne et al. (2018) report on small-scale experiments of passenger micro-behaviours during aircraft boarding and deplaning procedures. Their study suggested that the impact of luggage and seat pitch had a notable, but complex, effect on performance.

5.13. New measurement methods (e.g. pressure sensors)

A number of studies discussed new experimental angles in pedestrian dynamics. Nagao et al. (2018) developed a new measurement method of crowd density using machine learning. In their experiments, one pedestrian passed through a crowded area and they measured the body-rotational angular velocity with commercial tablets. By applying the continuous wavelet transform and machine learning to the data obtained from the experiment and showing that angular velocity has a relationship with crowd density, they estimated density. In a rather similar concept, Nicolas et al. (2019) explored mechanical response of dense pedestrian crowds to the crossing of intruders and identified both similarities and discrepancies with the response of granular media. Wang and Weng (2018) performed a series of controlled experiments on the collision dynamics in a queue of pedestrians and collected independent forces from five pressure sensors to restore the complete successive collision process. Wang and Weng (2018) reported on controlled experiments of the collision dynamics and the transmission pattern between pedestrians along the queue using sensors. Their observations suggested that the collision phenomenon will transmit along the queue of pedestrians continuously and unidirectionally, resembling the domino effect.

5.14. Social group behaviour

Group behaviour (Xie et al., 2020) was another matter of interest among studies of this category. Lu et al. (2017) proposed a floor field CA model to incorporate the pedestrian group behaviour based on the leader-follower rule. They calibrated the model with a few experiments of crowd evacuation conducted in a university building. Gorrini et al. (2016) also conducted experiment on dyads in a counter flow scenario. Subsequently, Crociani et al. (2018) proposed a model that accounts for the shape of groups in pedestrian simulation. Using experimental observations mentioned above, they performed calibration analyses on parameters governing the model for pedestrian groups. Köster et al. (2011) showed that the occurrence of groups significantly impacts crowd movement, particularly evacuation times. They suggested that with increasing group size, it becomes more difficult for groups to stay together at the door and the overall evacuation time increases. In the most recent experiment of social groups, Haghani et al. (2019e) investigated decision-making behaviour of social groups in evacuations by performing experiments under varying group sizes. They also observed that group size had a significant effect on both pre-movement and evacuation times, but the relationship between the group size and those two quantities was not monotonic. This topic was identified as another case of controversial topics in crowd dynamics and will be further discussed in the section dedicated to the controversial topics in Part II of this article.

5.15. Limited visibility conditions

Laboratory experiments under reduced visibility (Xie et al., 2020) was another major theme. Isobe et al. (2004) considered evacuation of a room without visibility and experimented it in a classroom. They observed that blindfolded students exhibited a distinctly different behaviour compared to a situation in which people can see their environment. Cao et al. (2015) also reported on experiments of evacuation without visibility and compared the experimental observations with that of a multi-grid model. In stark contrast with the finding of Isobe et al. (2004) who mentioned "adding more exits does not improve the situation in the expected way, since most people use the exit that is discovered first", Cao et al. (2015) suggested, based on their experiment, that "exit width has little impact on evacuation time and increasing number of exits is an effective way to decrease evacuation time". In a follow-up study, Cao et al. (2019b) further analysed their experiment and identified typical evacuation behaviours of occupants

including moving along walls as well as different conflict resolutions. This discrepancy of findings constitutes another major case of controversial topic in crowd dynamics. Another experiment of this category is reported in the study of [Cao et al. \(2019c\)](#) which was discussed already in the section related to the single-file experiments.

5.16. Pedestrians with limited mobility

Evacuation and motion of people with disability, as a major emerging topic identified in this review, was studied using laboratory experiments as well. A pioneering experiment of walking behaviour with mixed-ability individuals was conducted at Utah State University and has been reported and analysed in the studies of [Sharifi et al. \(2016\)](#), [Sharifi et al. \(2015\)](#), [Sharifi et al. \(2017\)](#) and [Stuart et al. \(2019\)](#). The experiment investigates walking behaviour of heterogeneous populations (including individuals with visual and mobility impairments) in various passageways including straight corridors, corners, bottlenecks and stairs. These publications collectively produced a range of useful findings and new insights into the walking behaviour of crowds with mixed ability of movement. Another experiment that they conducted in this domain, as reported in [Gaire et al. \(2018\)](#), investigated exit choice behaviour of pedestrians involving individuals with disabilities during building evacuations. The most recent study within this category is from [Georg et al. \(2019\)](#) who experimented the influence of wheelchair users on movement in a bottleneck and a corridor. They showed that the flow-density relation is significantly affected by the presence of individuals with disabilities. They observed an adaption of the overall movement speeds to the movement speeds of participants using a wheelchair, even for low densities and free flow scenarios.

As mentioned earlier, studies based on laboratory crowd experiments were many, and as a result, only central topics and the common themes in the preceding paragraphs were reviewed, which only covers nearly half of the total studies in this category. There were for example several experiments on route choice and they were mostly conducted for model calibration purposes ([Crociani et al., 2016](#); [Guo et al., 2013](#); [Werberich et al., 2015](#)). To keep this section at a reasonable length, the discussion on other individual studies is skipped and the readers are instead referred to [Table 3](#) for the complete list and details of these studies. Also a series of lab experiments conducted to explore the effect of the urgency/stress level on various aspects of evacuation behaviour ([Haghani and Sarvi, 2019a](#); [Haghani et al., 2020](#); [Haghani et al., 2019e](#)) will be discussed to in the following section when comparing their findings with those of the counterpart VR/AR studies.

6. Review of virtual/augmented reality and hypothetical choice experiments

The use of virtual reality (VR) and augmented reality (AR) methods in crowd and evacuation research was identified as one of the popular methods in this field that has generated a great portion of our knowledge in this area. This is a flexible method that can offer great levels of realism while allowing experimenters to enjoy the benefits of controllability in laboratory settings. The method appears to have maintained its popularity and nearly forty studies using this method were identified in this updated review. One may say that, given the limitations of staging laboratory crowd experiments, VR/AR experiments are crucial methods to study various aspects of crowd research particularly in the domain of emergency evacuations. While the studies of this category are dominated by AR/VR experiments, consistent with the previous review, questionnaire-based stated-choice experimental studies were accommodated within this category ([Chen et al., 2018e](#); [Shiwakoti et al., 2018](#); [Shiwakoti et al., 2017](#); [Shiwakoti et al., 2019b, 2020](#)).

Among the new studies that are reviewed in this work, a number of common themes were identified, some of which are traditional topics, but some are relatively new topics in experiments of this kind. These common themes include VR experiments of terror attacks ([Chittaro and](#)

[Sioni, 2015](#)), high-stress versus low-stress evacuation contexts ([Crescentini et al., 2016](#); [Lin et al., 2019](#); [Moussaïd et al., 2018](#)), evacuation training/education and serious gaming ([Almeida et al., 2017](#); [Aoki et al., 2007](#); [Crescentini et al., 2016](#); [Delcea and Cofas, 2019](#); [Kinatader et al., 2013](#); [Lovreglio et al., 2018](#); [Smith and Ericson, 2009](#)), wayfinding and spatial navigation ([Arias et al., 2019b](#); [Burigat and Chittaro, 2016](#); [Cao et al., 2019a](#); [Lin et al., 2019](#); [Meng and Zhang, 2014](#)), exit and route choice experiments ([Aleksandrov et al., 2018](#); [Chen et al., 2018e](#); [Kinatader et al., 2018](#); [Lovreglio et al., 2016a](#)), exit sign and evacuation guidance ([Duarte et al., 2014](#)), experiments of pre-evacuation activities (e.g. altruism, pre-movement time) ([Bode and Codling, 2018](#); [Gamberini et al., 2015](#); [Van den Berg et al., 2018](#)), evacuations of specific environments (e.g. aircrafts and tunnels ([Cosma et al., 2016](#))), and finally as an important topic that seems to be gaining momentum recently, the external validity of VR experiments ([Kinatader and Warren, 2016](#); [Li et al., 2019a](#); [Moussaïd et al., 2018](#)). The full details of these studies are synthesised in [Table 4](#).

6.1. Emotional, psychological and physiological response in VR evacuations (e.g. the stress effect)

[Chittaro and Sioni \(2015\)](#) conducted a comparative study of the interactive versus non-interactive terror attack simulation in a virtual train station in the form of serious gaming. They observed that “both versions of the simulation can provide positive outcomes in learning, risk severity perception and self-efficacy, but they differ in how much they affect user’s threat appraisal and emotional response”. The experiments reported by [Tucker et al. \(2018\)](#) tested the effect of hazard level (the presence of visible fire and smoke) on evacuees’ anxiety level and choice of egress route. Their results showed certain behavioural differences under the two different hazard levels. For example, the analysis suggested that “participants in the low hazard condition were significantly more likely than those in the high hazard condition to opt for peripheral exits over main exits”, and that “there was no significant effect of hazard on evacuation time”. Participants’ physiological responses like heartbeat rate showed an increase in the high hazard condition, as indicators of higher degrees of anxiety. The experiment of [Crescentini et al. \(2016\)](#) studied the impact of a mindfulness-oriented meditation intervention on the psychological and physiological responses evoked by four immersive virtual environments that were designed to elicit different levels of stress. These experiments were visualised in virtual trains, train stations and school buildings. The authors found out that the mindfulness meditation group showed lesser anxiety levels compared to the control group. [Lin et al. \(2019\)](#) studied the influence of repeated exposures and mental stress on human wayfinding performance in indoor environments. Their results showed significant positive impact of repeated exposure on participants’ wayfinding performance as well as significant negative impact of mental stress caused by the fire emergency on participants’ wayfinding performance. Wayfinding and navigation persist to be a popular topic in the VR/AR experiments. [Meng and Zhang \(2014\)](#) conducted a fire-emergency way-finding experiment in a simulated hotel building in which two different treatments were examined: without virtual fire and with virtual fire being presented to the participants. Their observations showed that participants that underwent the treatment condition (i.e. with fire) showed higher physiological and psychological stress, and their escape behaviour was influenced compared to the control group (e.g. took longer to find the exit signs and the exits). Using shared virtual-reality experiments in which multiple subjects interact with one another in the shared environment, [Moussaïd et al. \(2018\)](#) compared collective behaviours between non-emergency and stressful emergency conditions and their results revealed significant differences between the two conditions. In the non-emergency condition, participants searched relatively slowly for the exits and kept reasonably safe distance from one another. In contrast, severe crowding and higher numbers of collisions occurred during the stressful treatment. Another pattern reported

Table 3
Studies based on laboratory crowd experiments with real human participants in chronological order.

Author(s) (Year)	No. of subjects	Data extraction method	Physical setup	Density level	Speed level	Investigated phenomena	Measured variables	Design factors	Main observations or applications
Isobe et al. (2004)	Up to 10	Manual video analysis	Classroom	Solo to medium	Slow	Evacuation from smoky room	- Escape times	- Crowding level	- Disoriented students exhibited a distinctly different behavior compared to people who can see their environment - Adding more exits did not improve escape time, since most people used the exit discovered first - Pedestrians inside bottlenecks effectively form layers or trails that overlap (the Zipper effect) - The zipper effect causes the capacity of the bottleneck to increase in a stepwise fashion with the width
Hoogendoorn and Daamen (2005)	60 to 90	Image processing	Bottleneck	High	Slow	Bottleneck flow	- Headways - Speed	- Bottleneck width	- Under high-density conditions upstream of a bottleneck, the maximum capacity can drop - Controlling the population density upstream of bottleneck can maximise the flow through it (i.e. avoid capacity drop) - Improving the friction parameter - Introducing the cost of turning at exit - Validation using experiments - Pedestrian flow increased by placing an obstacle - Comparison between Chinese and French pedestrians - Average relaxation time and mean free speed were both larger for Chinese pedestrians - The force from the wall larger for French pedestrians - More obvious right preference in Chinese pedestrians
Cepolina (2009)	47	Manual video analysis	Bottleneck	High	Slow & fast	Bottleneck flow under oversaturated condition	- Egress flow rate	- Bottleneck width - Walking speed	- An obstacle in front of an exit in a congested situation increases the pedestrian outflow - Groups have a tendency to walk abreast to ease communication - Groups significantly impact crowd movement, namely evacuation times - There was a positive relationship between group size and egress time
Yanagisawa et al. (2009)	18	Manual video analysis	Narrow doorway	High	Slow	Bottleneck flow	- Flow rate	- Orderly vs disorderly egress behaviour	- Step length is proportional to the velocity of the pedestrian, and is directly related to the space available in front of him - Local density has a direct influence on the stepping characteristics - Synchronization of steps is particularly important at high densities - A rhythm that is slower than the normal-walking pace in free-flow situation increases flow in congested situation - Deriving a fundamental diagram for stairs downwards - Studying the influence of the slope of stair on FD - Comparison with a field study - Comparison with numerical simulations
Ma et al. (2010)	20	Image processing	Straight corridor	Low	Slow	Pedestrian flow in built corridor	- Speed - Wall force - Interactions	- Walk solo or in conflict	- Calibrating model parameters
Yanagisawa et al. (2010)	50	Manual video analysis	Narrow doorway	High	Slow	Bottleneck	- Flow rate	- Obstacle presence - Egress behaviour - Group size	
Köster et al. (2011)	30	Manual video analysis	Classroom and corridor	High	Slow	Group behaviour	- Evacuation time		
Jelić et al. (2012)	Up to 28	Automated motion capture	Ring corridor	Medium to high	Slow	Stepping behaviour	- Step length - Step duration - Velocity - Density	- Density level	
Yanagisawa et al. (2012)	Up to 30	Manual video analysis	Ring corridor	Medium to high	Slow	Unidirectional flow	- Flow - Density	- Density level	
Burghardt et al. (2013)	350	Digital image processing	Staircase	High	Slow	fundamental diagram of stairs	- Age - Navigation accuracy - time duration - Exit selection - Evacuation time - Egress rate	- Entrance width - Incoming flow - Density level - Initial positions - None	
Guo et al. (2013)	Up to 30	Manual video analysis	Classroom	Low to High	Slow	Route choice			
Hrabák et al. (2013)	Up to 30	Manual video analysis	Narrow doorway	High	Slow	Bottleneck flow			

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Table 3 (continued)

Author(s) (Year)	No. of subjects	Data extraction method	Physical setup	Density level	Speed level	Investigated phenomena	Measured variables	Design factors	Main observations or applications
Jiang et al. (2014)	76	Manual video analysis	Narrow doorway	High	Fast	Bottleneck flow	- Escape rate	- No. of obstacles	- Placing two pillars on both sides but not in front of the door can maximize the escape efficiency - Making comparisons with numerical simulation result - Comparison between simulation and experimental results - Calibrating model parameters
Shimura et al. (2014)	25	Manual video analysis	Straight corridor	High	Slow	Overtaking behaviour in mixed velocity flows (proxy for mixed age)	- Travel time	- Initial configuration	- Typical characteristics of blind evacuation, including the preference of choosing left-hand side direction and following behaviour, are found
Wolinski et al. (2014)	6	Automated image processing	Circle antipode	Low	Slow	Collision avoidance	- Trajectory	- Not reported	- Different strategies of conflict resolution are observed
Cao et al. (2015)	30	Manual video analysis	Square room with one exit	Solo to high	Slow	Blind evacuation	- Behaviour types - Velocity - Evacuation time	- Crowding level - Exit width	- Defining values of passenger saturation flows for different characteristics of public transport doors - Exploring traffic flow characteristic of heterogeneous pedestrian stream in various walking facilities - Calibration of a route choice model
Fernández et al. (2015)	26 and 50	Manual video analysis	Mock-up train/bus vehicle	High	Slow	Bottleneck flow	- Saturation flow	- Platform height - Door width - Flow direction - Flow composition	- Investigating crowd motion properties with different age compositions
Sharifi et al. (2015)	342	Automated image processing	Circuit and stairs	High	Slow	Traffic flow of pedestrians with mixed (dis)ability	- Flow - Density - Speed	- None	- Traffic jams occur more frequently in mixed group due to the great differences of mobilities and self-adaptive abilities
Werberich et al. (2015)	40	Automated image processing	Simple maze	High	Slow	Route choice	- Travel time - Travel distance - Route choice	- Choice set configuration	- Other pedestrians are generally perceived as repulsive - Choice of route is generally aimed at avoiding congestion
Cao et al. (2016)	127	Automated image processing	Closed ring corridor	Medium to high	Slow	Single file movement	- Flow - Speed - Density - Headway	- Age composition - Density level	- Calibration and validation of a simulation model - Measuring behavioural cohesion in individuals during response and movement phases of evacuation
Crociani et al. (2016)	46	Automated image processing	Rectangular arena with barricades	High	Slow	Route choice	- Choice	- Alarm type - Naivety - Crowding level	- Presenting two quantitative methods for assessing the collective behaviour in evacuees
Questa et al. (2016)	Up to 75	Manual video analysis	Transportation	Medium to High	Slow	Collective behaviour of evacuees	- Exit use - Response time - Exit time	- Entrance location - Obstacle	- Non-trivial patterns of space acquisition, e.g., an apparent preference for positions near corners, were observed
Ezaki et al. (2016)	25	Automated image processing	Square arena	High	Slow	Inflow process	- Density - Location preference - Walked distance - Flow - Density - Speed	- Flow ratio	- To understand lane formation, five different phases need to be distinguished - Once lanes are formed, the balanced bidirectional flow is the most efficient - Faster is slower
Feliciani and Nishinari (2016)	Not reported	Automated image processing	Straight corridor	High	Slow	Bidirectional streams	- Passage time - Burst sizes	- Competitiveness - Door width	- Distribution of burst sizes decays exponentially - Flow dynamics depends on the crowd size - Indeed, faster-is-slower is only observed in the central part of the evacuation
Garcimartín et al. (2016)	85 and 95	Automated image processing	Narrow doorway	Extreme	Slow to fast	Bottleneck flow	- Speed - Burst sizes	- Flow ratio - Group composition	- Increase of flow ratio negatively impacted the speed of pedestrians - Dyads walked significantly slower than singletons
Gorini et al. (2016)	54	Automated image processing	Straight corridor	High	Slow	Walking behaviour of groups (in counter flow)	- Speed	- Flow direction	
Guo et al. (2016)	Up to 100		Ring corridor		Slow		- Flow		

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Table 3 (continued)

Author(s) (Year)	No. of subjects	Data extraction method	Physical setup	Density level	Speed level	Investigated phenomena	Measured variables	Design factors	Main observations or applications
Liao et al. (2016)	250 and 300	Manual video analysis	Bottleneck	Medium to high	Slow	Directional flow under limited view	- Density	- View condition - Crowding level	- Flow rate under the view-limited condition decreases - Spontaneous lane formation did not occur under the view-limited condition - Pedestrian flow does not evolve into stationary state under view-limited and no-previous-experience condition - Proposing a method to detect steady states from bottleneck experiments
Nikolić et al. (2016)	Not reported	Automated image processing	Narrow bottleneck	High	Slow	Bottleneck flow	- Flow - Density	- Bottleneck width - Bottleneck length - Passage width - Crowd size - Not reported	
Parisi et al. (2016)	20	Automated image processing	Rectangular arena	Low to high	Slow	Collision avoidance	- Trajectory - Density	- Flowrate - Group size - Conflict configuration	- Proposing a probabilistic modelling approach to represent the speed-density relationship of pedestrian traffic - Investigating some features - of pedestrians' avoidance mechanisms in crossing and head-on encounters for different flow rates and group sizes
Sharifi et al. (2016)	231	Automated image processing	Circuit	High	Slow	Traffic flow of pedestrians with mixed (dis)ability	- Capacity - Headway - Speed - Spacing	- Flow direction - Flow composition	- Identifying the impact of involving individuals with disabilities on capacity estimation - Visually impaired and nonmotorized wheelchair users caused the minimum capacity reduction, whereas individuals using motorized wheelchairs caused the maximum capacity reduction
Colombi and Scianna (2017)	150	Field recording	Rectangular arena with non-walkable flowerbed in the middle	Medium	Slow	Route choice	- Social influence	- Choice	- Showing how different perceptions of the surrounding individuals give rise to completely different pedestrian dynamics - Prompting to modify Helbing social force model
Garcimartín et al. (2017)	55	Automated image processing	Narrow doorway	Extreme	Slow to fast	Bottleneck flow	- Density - Velocity - Kinetic stress field	- Competitiveness	- When sustained contact among the individuals exists, density is not enough to describe the evacuation
Guo et al. (2017)	50 and 100	Automated colour recognition tracking	Room with one exit	High	Fast	Bottleneck flow	- Evacuation time	- None	- The average evacuation time gap is constant.
Hänseler et al. (2017)	Up to 90	Manual video analysis	Conflicting streams	High	Slow	Multi-directional flow	- Time gaps - Walking time	- Conflict type - Geometry - Group size	- Testing a macroscopic loading model for multi-directional, time-varying and congested pedestrian flows
Lian et al. (2017)	Up to 295	Automated image processing	Merging corridors	High	Slow	Merging flows	- Flow - Density - Speed	- Width of the branch	- Lane formation was found downstream in the main channel - The main and branch channel became mutual bottlenecks to each other
Liao et al. (2017b)	350	Automated image processing	Bottleneck	High	Slow	Bottleneck flow	- Flow - Density - Velocity	- Bottleneck width	- Density in the corner increased with branch width - Proposing and testing a model validation procedure
Liu et al. (2017)	11	Automated image processing	Marked passage in open area	Solo	Slow	Direction changing	- Relaxation time	- None	- Pedestrians are insensitive to tiny angular difference in freely walking spaces - The relaxation time and free moving speed respectively were 0.659 s and 1.540 m/s - The relaxation time is not an invariable parameter, it tends to increase with an increase in the angular difference
Lu et al. (2017)	44	Manual video analysis	University building	High	Fast	Group behaviour in evacuation	- Speed - Evacuation time	- Group composition	- Validating a group behaviour model based on the concept of leader-follower behaviour (continued on next page)

Table 3 (continued)

Author(s) (Year)	No. of subjects	Data extraction method	Physical setup	Density level	Speed level	Investigated phenomena	Measured variables	Design factors	Main observations or applications
Rahman et al. (2017)	60	Automated image processing	Angled corridor	Medium	Slow	Walking speed	- Speed	- Turning angle	- Evacuation time significantly increases with the presence of pedestrian groups
Roy et al. (2017)	2	Not reported	Open arena	Duo	Slow	Conflict avoidance	- Trajectory	- Conflict configuration	- Estimating walking speed in various turning flow configurations
Seriani et al. (2017)	110	Manual video analysis	Metro train and platform mock-up	High	Slow	Boarding and alighting	- Density - Passengers distance	- Load condition	- Modelling pedestrians' avoidance dynamics based on a Fokker-Planck Nash game
Shahhoseini et al. (2017)	150	Automated image processing	Merging corridors	High	Slow to Fast	Merging flows	- Speed - Flow	- Merge angle - Merge configuration - Speed	- Interaction between the two players modelled as a static game, a static Nash game
Sharif et al. (2017)	311	Automated image processing	Circuit and stairs	High	Slow	Traffic flow of pedestrians with mixed (dis)ability	- Flow - Density	- Flow direction - Flow composition	- Numerical Nash equilibria compared to the output of real experiments
Sieben et al. (2017)	270	Automated image processing	Concert hall entry	High	Slow	Collective phenomena	- Speed - Density - Waiting time	- Entrance physical structure	- Proposing a new method to classify the interaction between passengers boarding and alighting
Xue et al. (2017)	80	Manual video analysis	Experiment in discrete space and time (a channel)	High	Slow	Pedestrian counter flow	- Step choice	- Channel width - Crowding level	- Delay is dramatically greater in asymmetrical setups than symmetric setups
Yuan et al. (2017)	50	Automated image processing	Open arena	Duo	Slow	Collision avoidance	- Avoidance distance - Direction choice	- None	- The difference is more pronounced at elevated levels of pedestrians' desired speed
Zhao et al. (2017)	Up to 70, 80, 45	Automated image processing	Circular ring + bottleneck + School corridor	Low to High	Slow to high	Self-slowing behaviour	- Headway - Velocity	- Crowding level - Normal v - emergency (speed level) - Panel obstacle	- An overview of conducting experimental research on pedestrian walking behaviour involving individuals with and without disabilities
Zhao and Zhang (2017)	Up to 70	Automated image processing	Circular ring	Low to high	Slow	Follow the leader	- Flow - Density - Spacing - Velocity	- Crowding level	- Showing the influence of the spatial structure on dynamics of the entrance procedure
Adrian et al. (2018)	Up to 75	Automated image processing	Corridor and entrance	High to extreme	Slow to fast	Entrance flow	- Density - Waiting time	- Corridor width - Crowd size - Motivation level	- With guiding barriers, density in front of the entrance was smaller
Bukáček et al. (2018)	Up to 76	Automated image processing	Narrow passage	Low to high	Slow	Bottleneck flow	- Travel time - Exit strategies	- Crowding level	- Collaboration behaviours of pedestrians were observed, enabling the formation of exit rows and prevent the formation of gridlock

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Table 3 (continued)

Author(s) (Year)	No. of subjects	Data extraction method	Physical setup	Density level	Speed level	Investigated phenomena	Measured variables	Design factors	Main observations or applications
Cao et al. (2018b)	350, 2000, 364	Automated image processing	Straight and crossing corridors	High to extreme	Slow to fast	Fundamental diagram	- Flow - Density - Velocity	- Crowding level - Physical configuration of movement	- Some pedestrians are able to push effectively through the crowd, some get trapped in the crowd for significantly longer time - The ability to push through the crowd is a combination of aggressiveness in conflicts and willingness to overtake - The difference of fundamental diagrams in pedestrian flow is investigated - Comparison of different measurement methods - Discrepancy of motivation and competitiveness in different experiments leads to large differences in FD
Cao et al. (2018c)	127	Automated image processing	Ring shaped corridor	High	Slow	Stepping behaviour	- Step length - Step width - Stepping time - Velocity - Gender - Height - Speed - Spatial pattern	- Age composition	- Investigating stepping behaviour of pedestrians with different age compositions
Crociani et al. (2018)	54	Automated image processing	Straight corridor	High	Slow	Walking behaviour of groups (in counter flow)	- Trajectories - Choice	- Flow ratio - Group composition	- Validating numerical results based on empirical data from a controlled experiment
Dias and Lovreglio (2018) Gaire et al. (2018)	Not reported 47	Automated image processing Radiofrequency identification	90° bend University building	Solo High	Slow Slow	Pedestrian manoeuvre around bends Exit choice in evacuation of people with disabilities	- Choice	- None - Participant distribution - Exit availability - Evacuation strategy	- Calibrating a CA model - The continuous representation fits the data better - Significant differences in exit choice between people with/out disability - Visual signs are of little importance to individuals with visual disabilities
Garcimartín et al. (2018)	Up to 179	Automated image processing	Narrow doorway	Extreme	Fast	Bottleneck flow	- Evacuation times - Time lapses - Velocities	- Obstacle - Obstacle position - Group size - Competitiveness - Door width - Luggage presence - Seat pitch - Instructions	- Pedestrians flow rate is not necessarily altered by the presence of an obstacle - In competitive evacuations, an obstacle attenuates collective transversal rushes, which are hazardous and can cause falls. - The impact of luggage and seat pitch had notable, but complex effect on performance
Gwynne et al. (2018)	35	Manual video analysis	Aircraft model	High	Slow	Aircraft boarding and deplaning	- Bag stow time - Bag collect time - Belt (un)fasten time - Brace time - Leave seat time - Speed - Choice	- Exit position - Exit width - Crowding level	- Investigating hypothetical bias in stated choice experiments of exit choice - Investigating the effect of decision rule specification in modelling exit choice data
Haghani and Sarvi (2018b)	Up to 142	Automated image processing	Multi-exit room	High to extreme	Slow to fast	Exit choice	- Evacuation time - Evacuation order - Aisle conflicts	- Aisle width - Evacuation direction	- Total evacuation time and the average rate decrease with increasing aisle width - Evacuation order of each occupant is mainly related to the seat position
Huang et al. (2018a)	25	Manual video analysis	Classroom	High	Slow	Evacuation in narrow aisle	- Speed - Headway - Density - Flow rate	- Aisle width - Walking speed - Inactive pedestrians - None	- Analysing the interferences of inactive pedestrians - Proposing a CA model in which the maximum density of crowd can reach to 8p/m ² - Verifying model reliability by experiments
Huang et al. (2018b)	25	Manual video analysis	Classroom	High	Slow & fast	Single-file movement in narrow aisle	- Evacuation time - Evacuation order - Aisle conflicts	- Aisle width - Walking speed - Inactive pedestrians - None	- Analysing the interferences of inactive pedestrians - Proposing a CA model in which the maximum density of crowd can reach to 8p/m ² - Verifying model reliability by experiments
Ji et al. (2018a)	943	Manual video analysis	School building	High	Slow	Pedestrian evacuation	- Evacuation time - Flow rate	- Aisle width - Walking speed - Inactive pedestrians - None	- Analysing the interferences of inactive pedestrians - Proposing a CA model in which the maximum density of crowd can reach to 8p/m ² - Verifying model reliability by experiments

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Table 3 (continued)

Author(s) (Year)	No. of subjects	Data extraction method	Physical setup	Density level	Speed level	Investigated phenomena	Measured variables	Design factors	Main observations or applications
Jin et al. (2018)	Up to 235	Automated image processing	Ring corridor	Extreme	Slow	Unidirectional flow	- Density - Behaviour types	- Crowding level	- Showing experimentally that pedestrian density can in fact reach $8p/m^2$ - Global density as high as $9p/m^2$ is observed. - Flow dynamics in hyper-congested and over-congested states are different - The critical density is $8p/m^2$ - The critical density between over-congested and congested states is $6p/m^2$ - Identifying different types of individual behaviour - Measuring walking speed on an inclined corridor as a model of sinking ship
Kim et al. (2018)	2	Manual video analysis	Inclined corridor	Solo	Slow	Evacuation from sinking ship (walking speed)	- Speed	- None	- The break of spatial configuration of people in the room due to new entries - The repulsion effect
Lian et al. (2018a)	Up to 40	Automated image processing	Rectangular room	High	Slow	Inflow into a room	- Spatial patterns	- (In)active persons	- The mean maximum distance increases with the increase of the number of inactive persons
Lian et al. (2018b)	Up to 40	Automated image processing	Rectangular room	High	Slow	Inflow into a room	- Spatial patterns - Distances	- (In)active persons	- Proposing a video data-driven social force model
Liu et al. (2018)	Not reported	Automated image processing	School building	High	Slow	Crowd evacuation	- Trajectories	- None	- Initialization of pedestrian position, path navigation, and goal selection in the improved social force model is guided by real video data
Ma et al. (2018)	Up to 70	Automated image processing	Ring corridor	Low to High	Slow	Stepping behaviour in single-file movement	- Step length - Step duration - Velocity - Spatial headway	- Crowding level	- The relationship between step length (duration) and spatial headway exhibits piecewise linear behaviour - The relationship between step duration and step velocity is non-monotonic - The relationship between step length and step velocity is monotonic
Nagao et al. (2018)	Up to 40	Gyroscopic sensor	Square arena	Low to high	Slow	Estimation of crowd density	- Density - Angular velocity	- Crowding level	- Developing a new measuring method of crowd density using machine learning - Wavelet transform improves accuracy crowd density prediction - Angular velocity is related to crowd density and can be used to estimate crowd density
Seriani and Fujiyama (2018)	110	Manual video analysis	Mock-up of a metro carriage & platform	High	Slow	Boarding and alighting	- Spatial pattern	- Boarding and alighting ratio	- Passenger space for alighting passengers can be represented as an asymmetrical ellipse
Shahhoseini and Sarvi (2018)	150	Automated image processing	Merging corridors	High	Slow to fast	Merging flows	- Evacuation times	- Setup symmetry - Movement speed - Merging angle - Movement speed	- Validating numerical simulation
Shahhoseini et al. (2018)	150	Automated image processing	Merging corridors	High	Slow to fast	Faster is slower	- Flow rate - Time lapses	- Movement speed	- Faster is not slower
Song et al. (2018a)	26	Automated image processing	Room & corridor	High	Slow to fast	Evacuation	- Trajectories - Speed - Density	- Competitiveness - Obstacle position - Crowding level - Door width - Door width - Competitiveness	- Proposing a multi-scenario adaptive neural network to model pedestrian behaviour
Song et al. (2018b)	50	Manual video analysis	University class	High	Slow to fast	Evacuation	- Evacuation time - Flow rate	- Door width - Competitiveness	- Calibrate the tangential characteristic time in social force model to avoid hitting the wall
Sun et al. (2018)	6	Electrical monitoring system	Mock-up of ship corridor	Medium	Fast	Single-file movement (on ship corridor)	- Speed - Density - Headway	- Trim angle - Heeling angle	- Nonlinear exit capacity model is proposed - Not only density, but also trim or heeling angles affect the speed - The front neighbour of the target subject has more influence than others in front

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Table 3 (continued)

Author(s) (Year)	No. of subjects	Data extraction method	Physical setup	Density level	Speed level	Investigated phenomena	Measured variables	Design factors	Main observations or applications
Templeton et al. (2018)	280	Manual video analysis	Open campus area	High	Slow	Social identity (psychological crowds)	- Distance - Speed - Spatial pattern	- Social identity - Priming	- Longer time headway needed in lower walking speed to keep enough headway - Heeling angles show less impact on speed when compared with trim angles. - Investigating behavioural effects of shared social identity on crowd movement - Compared to physical crowd, members of psychological crowd walk slower, walk further, and stay closer
Wang and Weng (2018)	6	Pressure sensors	Corridor & staircase	Medium	N.A.	Collision dynamics in a queue	- Collision force	- Surface type - Interpersonal distance	- Collision will transmit along the queue continuously and unidirectionally, resembling the domino effect
Wang et al. (2018b)	Up to 29	Automated image processing	Ring corridor	Low to high	Slow	Single-file movement	- Step length - Step frequency - Swaying amplitude	- Crowding level	- Linking pedestrian flow characteristics with stepping locomotion - Different types of step adaptation under congested conditions are observed
Wang et al. (2018c)	Up to 30	Automated image processing	Ring corridor	Low to high	Slow	Stepping locomotion (in single-file movement)	- Speed - Density - Step length	- Crowding level	- Six different stepping styles are observed - The proportion of stepping styles changes with density
Zeng et al. (2018)	Up to 60	Automated image processing	Ring corridor	Low to high	Slow	Stepping behaviour (in single-file movement)	- Step length - Step frequency - Density	- Crowding level	- Investigating distribution of step length and frequency - Identifying the relationship between step length and frequency under different headway
Zhang et al. (2018a)	25	Manual video analysis	Straight corridor	High	Slow	Overtaking behaviour in mixed velocity flows (proxy for mixed age)	- Travel time	- Initial configuration - Speed composition	- Step length and frequency increases with density - The effects of the initial pedestrian formation on overtaking behaviour and the evacuation time analysed
Bode et al. (2019)	319 & 603	Automated image processing	Intersecting corridor	High	Slow	Intersecting streams	- Flow - Density	- No. of streams - Roundabout - Column - Entrance width (inflow rate) - Exit width - Exit location - Density level	- Pedestrian flow influenced by both the initial positions and density of the slow pedestrians - As arrival rates increase, streams start to interact and compete for space - These interactions result in an even balance across the two intersecting streams
Cao et al. (2019b)	30	Automated image processing	Rectangular room	Solo & High	Slow	Evacuation under no visibility	- Individual evacuation time - Movement distance - Distance along walls	- Neither of the streams can dominate - Typical evacuation behaviour including moving along walls and different conflict resolutions are observed	
Cao et al. (2019c)	Up to 30	Automated image processing	Ring corridor	Low to high	Slow	Single-file movement (under limited visibility)	- Velocity - Headway - Density - Trajectories	- Visibility (light transmission) - Crowding level	- Typical behaviour of following the boundary and frontal pedestrian are observed under poor visibility - Pedestrians are more sensitive to the density change under good visibility
Georg et al. (2019)	Up to 92	Automated image processing	Bottleneck & narrow corridor	High	Slow	Bottleneck and corridor flow (of mixed ability crowd)	- Flow - Density - Speed	- Physical setup - Bottleneck/corridor width - Crowd composition - Exit availability	- The fundamental diagram is significantly affected by the presence of people with disabilities
Haghani and Sarvi (2019b)	114	Automated image processing	Rooms and corridors	High	Fast	Initiative (herd) behaviour (in direction choice)	- Evacuation time	- Initiative behaviour in direction choice is determinantal to evacuations - Mis-specifying herd tendency significantly corrupts simulation outputs	

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Table 3 (continued)

Author(s) (Year)	No. of subjects	Data extraction method	Physical setup	Density level	Speed level	Investigated phenomena	Measured variables	Design factors	Main observations or applications
Haghani and Sarvi (2019a)	Up to 114	Automated image processing	Rooms and corridors	Medium to High	Slow to fast	Direction choice	- Exit choice - Corridor choice	- Stress (urgency) level - Crowding level - Exit availability - Number of exits - Exit widths - Exit positions - Crowding level	- Herd behaviour is not likely in direction choice - Increased stress or crowding both make herd behaviour less likely
Haghani and Sarvi (2019e)	142 & 114	N.A.	Multi-exit rooms	Medium to high	Fast	Exit choice (changing)	- Evacuation times - Exit shares	- N.A.	- Decision changing is a critical feature of evacuation modelling - Extreme degrees of decision changing are not optimal - Intermediate degrees of imitation in decision changing is beneficial - Exit choice optimality is moderated by the level of crowdedness - Faster is not slower when there is no aggressive pushing
Haghani and Sarvi (2019d)	142 & 114	Automated image processing	Multi-exit rooms	High to extreme	Fast	Rationality in exit choice	- Evacuation time - Choice	- N.A.	
Haghani et al. (2019d)	114	Automated image processing	Narrow doorway	High to extreme	Slow to fast	Bottleneck flow	- Evacuation times - Headways - Burst sizes - Speed - Density	- Competitiveness - Door width	
Haghani et al. (2019e)	148	Automated image processing	Multi-exit room	High	Slow to fast	Social groups	- Reaction time - Decision time - Evacuation time - Exit choice - Exit choice changing - Density	- Stress level - Group size	- Relationship between group size and evacuation/ reaction time was not monotonic - Most group decisions were made through leadership - Groups showed imitation tendency in decision changing - Groups' exit choice can be approximated by individuals'
Hu et al. (2019)	72	Automated image processing	Circle antipode	Medium to high	Slow	Multi-directional (conflicting) flows	- Trajectories - Spatial displacement - Length of walk	- Crowding level	- Identifying typical pedestrian behaviours in multidirectional flows - Linear relationship between the length of walk and crowding level - More than 70% chose the straight strategy to reach destination - Pedestrians changing their strategies were observed - Luggage has little influence on the pedestrian walking speed - Luggage made average distance headway larger - Luggage reduced the acceleration rate - With luggage, the stable point of FD moved further towards the lower density side - Exploring the impact of obstacle size on the evading behaviour - Increasing obstacle width, did not affect the frequency and amplitude of body sway - By increasing obstacle width, the lateral deviation increases - Comparing fundamental diagrams across experiments - Critical density is about 3 ped/m
Huang et al. (2019)	26	Automated image processing	Straight passage	Solo to high	Slow	Luggage-laden pedestrian walking	- Speed - Acceleration - Spatial headway - Fundamental diagram	- Luggage type - Pedestrian composition - Individual v single-file walk	
Jia et al. (2019)	32	Automated image processing	Straight corridor	Solo	Slow	Obstacle evading behaviour	- Body sway - Stride length - Speed	- Obstacle width	
Jin et al. (2019a)	Up to 203	Automated image processing	Circular ring	High	Slow	Single-file movement	- Flow - Density - Velocity - Headway	- Crowding level	
Nicolas et al. (2019)	Up to 40	Automated image processing	Open arena	High to extreme	N.A.	Crossing of dense static crowds	- Spatial displacement - Density	- Intruder type (cylindrical v non-cylindrical) - Density level - Behavioural instruction	- Identifying similarities and discrepancies with response of granular media - The presence of a depleted region behind the intruder - Unlike grains, pedestrians anticipate intruder's passage by moving much before contact

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Table 3 (continued)

Author(s) (Year)	No. of subjects	Data extraction method	Physical setup	Density level	Speed level	Investigated phenomena	Measured variables	Design factors	Main observations or applications
Köster et al. (2019)	22 + 12	Stopwatch	Staircases	Solo to medium	Slow	Walking on stairs	- Speed	- Ascend v. descend - Density level	- Pedestrian interactions even at high densities do not become mechanical - Findings question the use of force models for dense crowds - Comparing free-flow speed of students in the plane and on stairs to investigate if there is a linear dependency - Investigating the effect of fatigue - Identifying different passenger behaviour depending on load conditions - Traffic efficiency is demonstrably sensitive to the physical features of the merging passages as well as the desired speed level of the crowd
Seriani et al. (2019)	110	Manual video analysis	Mock-up of a metro carriage & platform	High	Slow	Boarding and alighting	- Boarding/alighting time	- Boarding/alighting ratio	
Shahhoseini and Sarvi (2019)	160	Automated image processing	Merging corridors	High	Slow to fast	Merging flows	- Flow - Density - Egress time - Velocity - Headway	- Merging angle - Merging symmetry - Movement speed	
Shi et al. (2019)	50	Automated image processing	Bottleneck	High	Slow to fast	Bottleneck flow	- Headway - Flowrate - Exit time	- Centre v corner exit - Obstacle placement - Obstacle position - Obstacle size - Speed	- Obstacle has different performance at middle and corner exits - Column size and distance to the exit affect the outflow of pedestrians - 'Faster-is-faster' effect is observed under slow running condition
Stuart et al. (2019)	231	Automated image processing	Circuit	High	Slow	Traffic flow of pedestrians with mixed (dis)ability	- Capacity - Headway - Speed	- Flow direction - Flow composition	- Individuals with visual or stamina impairments are overtaken by other pedestrians at high rates
Tavana and Aghabayk (2019)	94	Automated image processing	Bottleneck	High	Slow	Bottleneck flow (efficient angel design)	- Spacing - Flow - Evacuation time - Headway - Velocity	- Funnel angle and shape	- Best-performing bottleneck with regard to angle follows a certain trend, with 26.6° being the most efficient
Wang et al. (2019a)	142	Automated image processing	Multi-exit room	High	Fast	Pedestrian movement dynamics	- Step choice	- No. of exits - Position of exits	- Adopting machine learning methods to study evacuees' stepwise movement - Consistency across machine learning and discrete choice modelling outcomes
Wang et al. (2019b)	Up to 916	Automated image processing	Corridors (straight & intersecting)	High to extreme	Slow	Fundamental diagram of pedestrian traffic	- Speed - Density - Flow	- Flow directions - Flow conflict type - Corridors width - Crowding level	- The fundamental diagrams for various flow types surveyed by using Voronoi diagram method
Xiao et al. (2019b)	Up to 64	Automated image processing	Circle antipode	Medium to high	Slow	Collision avoidance	- Route length - Route potential - Travel time	- Radius - Crowding level	- A model evaluation based on the experimental results is performed
Yamamoto et al. (2019)	4	Sensor	Narrow corridors	Low	Slow	(Body rotation in) collision avoidance	- Velocity - Angular velocity - Travel time	- Corridor width	- Pedestrians rotated their bodies when the corridor width was smaller than the sum of the widths of the two
Zeng et al. (2019)	Up to 40	Automated image processing	Ring corridor	Medium to high	Slow	(Effect of music on) single-file movement	- Flow - Density - Speed	- Rhythm v normal - Rhythm type - Crowding level	- Investigating the effect of background music on pedestrian movement - Velocity and flow with music at medium and high densities were smaller than those under normal conditions

in their study specific to the stressful treatment was the herding behaviour. They report that “the majority of participants herded in the same direction in the emergency condition, which amplified the crowding pattern”. Although in an earlier study published partially based on these experiments, the authors provided an argument indicating that this pattern may not necessarily be the result of an increased individual tendency to follow others and may rather be due to the increased crowdedness under the stressful treatment (Moussaïd et al., 2016).

The topic of evacuation behaviour under various levels of stress is of itself relatively new and one that has only gained attention within the last few years (Lin et al., 2020b). Unless we consider studies of bottleneck at different levels of competitiveness (Daamen and Hoogendoorn, 2012; Garcimartín et al., 2016; Haghani et al., 2019d) or studies with ants/mice under varying levels of aversive stimulus (as a proxy for the urgency level) in the same category of evacuation under varying levels of stress, studies on the topic of stress/urgency are mostly limited to the VR method. In 2017, however, a series of human crowd laboratory experiments was conducted under two different levels of stressful treatment. These studies investigated topics of herding in direction choice (Haghani and Sarvi, 2019a), exit choice, exit choice changing and reaction time (Haghani et al., 2020), as well as the reaction time, exit choice and exit choice changing of social groups (Haghani et al., 2019e). To contrast them closely with the counterpart VR experiments, a collective summary of their findings in relation to the effect of stress level on various aspects of evacuee behaviour is presented here. It was found out based on those experiments that stressful treatment has significant impact on the reaction time of individuals and groups, as well as their decision changing behaviour. Reaction times were shorter and people (regardless of whether they were in groups or as individuals) tended to change their exit choices more frequently under the higher urgency treatment (contrary to the previous observation of Bode and Codling (2013) regarding the exit choice changing in a VR setting). An overall observation was that evacuees initiated their movement more instantly and they moved faster under the high-stress treatments and this resulted in higher levels of crowdedness at exits and bottlenecks. This observation is particularly consistent with the study of Moussaïd et al. (2018). However, this increased level of rush and crowdedness did not result in prolonged evacuation times. To the opposite, despite increased densities at bottlenecks, evacuation times were invariably shorter under high-stress treatments. This aspect can be regarded to be at odds with the faster-is-slower assumption (Shahhoseini et al., 2018). It was also observed that under the high-stress treatment there is no increased tendency to follow direction choices of others (consistent with Bode and Codling (2013) and Haghani and Sarvi (2017a)), and if anything, the increased urgency level amplified the tendency to ‘not’ follow direction choices of others. This observation is in stark contrast with the suggestions of ant experiments of symmetry breaking. It was also observed that the effect of the stress level on exit-choice behaviour was rather negligible (to some degrees in contradiction with the observations of Tucker et al. (2018), although the two studies are not closely comparable). Readers can find a more detailed and more complete version of these contrasts in the section related to the controversial topics, as the topic of evacuation under stress appears in that list.

6.2. Altruism in VR evacuations

Gamberini et al. (2015) studied the effects of victim ethnicity and emergency type on helping behaviour. Their experiments focused on the withdrawal of help due to racial discrimination and explored the potential of virtual environments to trigger this bias in emergency situations. Their results indicated the presence of racial discrimination bias in help provision and also showed that the racial discrimination tendency increases under the time pressure treatment.

6.3. Evacuation training and education using VR

The topic of evacuation education and training is one that has received an unprecedented attention by the VR/AR studies. As will be demonstrated in following sections, this was in fact identified as one of the top emerging topics in the contemporary crowd dynamics literature. Kinateder et al. (2013) investigates the effect of simulated VR behavioural training on self-evacuation of road tunnels. They observed that the group trained in the VR setting evacuated a real road tunnel faster and more reliably compared to the control group. The study of Lovreglio et al. (2018) reports on prototyping a virtual reality serious game for building earthquake preparedness. Their experiment simulates an earthquake evacuation in a hospital building, as the case study. Delcea and Cotfas (2019) explored the possibility of using agent-based simulations in order to increase pupils’ awareness on how to act during an evacuation process caused by a fire. A post training classroom evacuation experiment showed a positive effect and an enhanced evacuation knowledge among students. In other studies, Smith and Ericson (2009) demonstrated how immersive game-based VR can be utilised to teach fire safety skills to children, and Aoki et al. (2007) employed the VR method for emergency egress navigation training from spacecrafts.

6.4. Spatial knowledge and navigation in VR evacuations

Wayfinding and navigation persist to be a popular topic in the VR/AR experiments (Lin et al., 2020a). Cao et al. (2019a) explored the influence of spatial exploration mode on people’s wayfinding performance during a building fire emergency. They asked participants to actively or passively explore a virtual museum during a virtual fire emergency. They observed that participants who explored the environment actively took longer to complete the egress task. Within a similar theme, Burigat and Chittaro (2016) propose a VR-based tool to support acquisition of spatial knowledge for evacuation purposes. In doing so, they also compared the effect of passive versus active navigation, and also compared those with a diagrammatic map. They observed that the VR method produces better spatial knowledge and that active navigation results in performance improvement.

6.5. Exit and route choice in VR evacuations

Exit and route choice behaviours are conventional topics investigated by VR/AR methods. The study of Chen et al. (2018e) reports on a questionnaire-based experiment of route choice with school children. Their findings showed that spatial position, congestion, group behaviour and backtracking behaviour have all significant effect on children’s route choice. The stated exit choice experiments of Lovreglio et al. (2016a) in the form of non-immersive VR choice scenarios suggested that “the presence of smoke, emergency lighting, distance of exit, number of evacuees near the exits and the decision-maker and flow of evacuees through the exits significantly affect local exit choice”. The experiments of Kinateder et al. (2018) also revealed that exit choice in an emergency evacuation scenario is influenced by exit familiarity and neighbour behaviour, jointly. They also showed that the social influence increases with the number of neighbours. Aleksandrov et al. (2018), using non-immersive VR experiments investigated the choice behaviour of evacuees in tall buildings. They observed that people tend to choose evacuation lifts when they are suggested as the main exit option, and when stairs are overcrowded. Despite variations of the findings from these individual studies on exit choice, one message has been consistently emerging from these experiments: exit choice behaviour is “jointly” influenced by a set of various factors including the social influence and the physical attributes of the exits. People make trade-off between these factors and their choices are not determined solely by a single factor, such as following others or choosing the nearest exit. This observation is also consistent with the body of exit choice studies that were surveyed in the previous review and can

therefore be regarded as one of the solid and well-established findings in the evacuation dynamics literature, reaffirmed time and time again by independent experiments (Bode et al., 2014, 2015; Bode and Codling, 2013; Haghani and Sarvi, 2016a–c, 2017b,c; Haghani et al., 2015a; Haghani et al., 2015b; Liao et al., 2017a; Lovreglio et al., 2014; Lovreglio et al., 2016b).

6.6. Pre-evacuation time in VR evacuations

Multiple VR experiments of pre-evacuation activities were published in the last few years. Using non-immersive VR experiments, Bode and Codling (2018) investigated the level of risk that individuals take by collecting virtual objects before evacuating. They observed that nearly 45% of participants chose extreme strategies by collecting either all or none of the available objects before evacuating. The study of Van den Berg et al. (2018) quantified the effect of social influence on the decision to evacuate, based on a serious game VR experiment. Their observations suggested that “the more people someone sees leaving, the more inclined this person is to leave” and that “seeing people leave has more impact than seeing people stay”.

6.7. Survey methods

A group of studies in this category, have used hypothetical-choice experiments but not with the aid of VR methods for their scenario presentation (Aleksandrov et al., 2018; Lovreglio et al., 2014; Lovreglio et al., 2016a), rather, in a more conventional form of hypothetical choice surveys. The choice surveys reported in (Shiwakoti et al., 2018) and (Shiwakoti et al., 2017) examined the likely behaviour of passengers in Australia under emergency scenarios in train stations (see Haghani and Sarvi (2016a) and Haghani and Sarvi (2017d) for more hypothetical choice surveys in the context of train station evacuation within the same geographical region). These surveys have indicated that people are more likely to be reactive and wait for the instructions from the authorities as opposed to initiating a self-evacuation and moving to exits. People also indicated in those surveys that they are more likely to behave cooperatively with others as opposed to pushing them out of their way. Also more recently, based on a survey of airline passengers at two airports (in Australia and China), Shiwakoti et al. (2020) and Shiwakoti et al. (2019b) examined the passengers' awareness and perception of emergency wayfinding tools. They found differences in the degree of emergency procedure awareness related to the geographical region of the respondents. Chen et al. (2018e) also used a questionnaire-based method to study the route choice behaviour of elementary school students in classrooms. Through this survey, they observed that the initial spatial position, congestion, group behaviour and backtracking have significant effects on children's route choice while the effect of gender and guidance were observed to be insignificant. Alnabulsi et al. (2018) administered a survey questionnaire to nearly 1200 pilgrims at Hajj, through which the link between shared social identity and giving support to others were examined. Other studies of this category include surveys of vulnerable building occupants, such as elderly or those with mobility impairments, to obtain their perspectives on evacuation methods (Butler et al., 2017) or to understand their fire risk perception (Tancogne-Dejean and Laclemece, 2016).

6.8. External validity of VR evacuations

An important question in regard to the applications of VR methods in crowd and evacuation research is the question of external validity: to what extent observations made in the VR/AR or hypothetical choice settings translate to the real-world behaviour (or at least the behaviour in more realistic settings, like laboratory crowd experiments). On this topic, a few studies were identified in the last review of the experimental literature all providing very promising evidence regarding the

applicability of the VR (or stated choice) methods to the real world (Haghani and Sarvi, 2017d; Haghani et al., 2016; Moussaïd et al., 2016). In the new review period, four more studies were identified to have addressed this topic (Haghani and Sarvi, 2018b, 2019c; Kinatader and Warren, 2016; Li et al., 2019a) whose collective finding, consistent with the previously reviewed studies, again points to the general idea that VR crowd experiments produce externally valid outcomes (to varying degrees). Kinatader and Warren (2016) compared the influence of bystander behaviour on the decision to evacuate of participants, in response to a fire alarm, in matched physical and virtual environments. Their outcomes showed that “the active bystander group was more likely to evacuate and the passive bystander group less likely to evacuate than the control group. This pattern of social influence was observed in both the real and virtual environments, although the overall response to the virtual alarm was reduced; positive influence of bystanders was comparable, whereas negative influence was weaker in VR”. Moussaïd et al. (2018) described two studies that have compared avoidance manoeuvres and simple evacuation situations against real-world data. Haghani and Sarvi (2018b) showed that exit choice models obtained from hypothetical choice experiments had great degrees of similarity with counterpart models derived from realistic (field-type) crowd evacuation experiments, and that the observation is valid irrespective of how we set the decision rule in the modelling phase. By implementing these empirically estimated choice models for prediction and integrating them with their crowd simulation software, they subsequently showed that the consistency between the two is even higher when considering aggregate simulation outcomes such as exit shares and evacuation times (Haghani and Sarvi, 2019c). Li et al. (2019a) also compared field and VR observations of route choice and found qualitative agreements across the outcomes of the two. More recently, Arias et al. (2019a) introduced the notion of supplementing fire witness's account with a virtual-reality forensic tool in order to generate knowledge about human behaviour in fires. To assess their methods, they replicated the conditions of a real hotel room fire and exposed naïve participants to it. A comparison of the data with that of the actual survivors of that fire demonstrated promising degrees of consistency, indicating that the behavioural response observed in the virtual fire experiment could be a reasonable proxy for human responses during real fire evacuation incidents.

7. Review of evacuation drill experiments

The set of the evacuation drill studies identified in this updated review is rather dominated by publications on the topic of tall building evacuation and stair evacuations (which is generally referred to as ‘vertical evacuation’ in this work). Nearly a dozen papers have addressed this topic using this experimental technique. As it will be shown in quantitative terms in a section dedicated to descriptive statistic in Part II of this article, the evacuation drill method has been one of the top two instrumental methods (along with the VR/AR experiments) in addressing the emerging research questions (i.e. questions that are gaining increasing attention) in crowd dynamics. Pedestrian movement on stairs is of itself identified as one of the major emerging topics in crowd dynamics. Among the studies of movement on stairs that use the evacuation drill method, three studies are in line with the rising attentions to the topic of evacuation of vulnerable people (Jiang et al., 2012; Kuligowski et al., 2013; Lam et al., 2014), three are related to the group behaviour (Chen et al., 2017a; Ma et al., 2017; Zhang et al., 2018d), and three are related to evacuation under limited visibility conditions (Chen et al., 2018c; Zeng et al., 2017; Zhang et al., 2018d) which themselves qualified as other emerging topics according to this review, as will be discussed later.

7.1. Evacuation drills with mixed ability crowds

Kuligowski et al. (2013) studied the stair descending of older adults

Table 4
Studies based on experiments using virtual/augmented reality or hypothetical choice methods in chronological order.

Author(s) (Year)	Method	Sample size	Hypothesised setup	Visualised (simulated) agents	Subject's task	Investigated phenomena	Design factors	Measured variables	Main observations or application
Aoki et al. (2007)	- 3D - Immersive - HM display	36	- Underground railway station	Crowd	- Navigate through virtual space station	- Navigation training for emergency egress from spacecraft	- Orientation - Smoke	- Egress time	- Individual 3D spatial abilities and orientation during training influence egress performance - Training enhances sense of direction and thereby, performance in low visibility - The promise of using game-based VR for fire safety skills training of children.
Smith and Ericson (2009)	- 3D - Immersive - CAVE - Game	21	- House	Solo	- Hazard search - Fire evacuation	- Fire response training	- Training	- Correct answers	- Trained participants showed better and faster behavioural responses than informed only participants - The validity of VR training consistent with field training
Kinatader et al. (2013)	- 3D - Immersive - HM display	43	- Road tunnel	Solo	- Drive and navigate (evacuate) through tunnel	- Self-evacuation in road tunnel accidents	- Information - Training	- Adequate reactions	- Dynamic sign produced the highest compliance, but the difference between dynamic and static was only significant for un-cued signs
Duarte et al. (2014)	- 3D - Immersive - HM display	90	- Company building	Solo	- Emergency egress	- Dynamic signage	- Signage type	- Behavioural compliance	- Salient (e.g., dynamic) signs are useful in benefiting compliance - Fire treatment condition induced higher stress, and influenced escape behaviour compared to the control group
Meng and Zhang (2014)	- 2D - Panorama screens	40	- Hotel building	Crowd	- Navigate	- Way-finding in fire emergency	- Presence of fire, smoke, alarm, explosions	- Skin conductivity - Heart rate - Eye movement - Pre-movement time - Escape time - Travel distance	- Both versions of the simulation provide positive outcomes in learning, risk severity perception and self-efficacy - They differ in how much they affect user's threat appraisal and emotional response.
Chittaro and Stoni (2015)	- 3D - Interactive - Immersive - Game	44	- Train station	Crowd	- Respond to terror attack scenario	- Interactive versus non-interactive serious game for terror response	- Interaction level	- Learning - Risk perception - Self-efficacy	- Bias by racial discrimination in help provision - Racial discrimination increases under time pressure, but not in a fire
Gamberini et al. (2015)	- 3D - Interactive - Immersive	96	- Building	Duo	- (Not) help the victim	- Helping behaviour and navigation	- The race of the virtual person requesting help	- Help provision - Time to react	- Calibrating floor field cellular automaton models
Lovreglio et al. (2015)	- 3D - Immersive - Interactive - CAVE	96	- Tunnel	Solo	- Navigate	- Navigation	- N.A.	- Exit choice	- Active navigation produces a performance improvement
Burigat and Chittaro (2016)	- 3D - Immersive - HM display	54	- Aircraft	Solo	- Navigate	- Acquiring spatial knowledge through VR	- Training type (Active v passive) - Visibility	- Usability measures - Evacuation time - Waiting time - Exit errors	- Positive impact of way-finding lighting installations on people - Evacuation safety - Decreased anxiety and lower heart rate in response to the simulated emergency for the meditation group
Cosma et al. (2016)	- 3D - Immersive - HM display	60	- Railway tunnel	Single	- Navigate	- Wayfinding in smoke-filled evacuation	- Lighting system	- Movement paths - Evacuation times	
Crescentini et al. (2016)	- 3D - Interactive - Immersive - HM display	41	- School building - Train station	Single	- Evacuate or explore building	- Stress level - Meditation training	- The effect of meditation on disaster response	- State anxiety - Trait anxiety - Physiological response	

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Table 4 (continued)

Author(s) (Year)	Method	Sample size	Hypothesised setup	Visualised (simulated) agents	Subject's task	Investigated phenomena	Design factors	Measured variables	Main observations or application
Kinatader and Warren (2016)	- 3D - Immersive - HM display	150	- Room	Solo/duo	- React to fire alarm	- VR validation - Peer influence	- Bystander presence/type	- Odds of leave/stay - Perceived risk	- Active bystander group was likelier to evacuate - Positive influence of bystander (social influence) observed in both the real and virtual setting - How static and dynamic information influence evacuation route choice
Lovreglio et al. (2016a)	- Non-Immersive - Online SP Survey	1503	- Room with two exits	Crowd	- Make choice	- Exit choice	- Smoke - Emergency lighting - Distance to exit - Social influence - Individual type	- Choice	
Tancogne-Dejean and Lacleme (2016)	- Interview	11	- Buildings	N.A.	- Answer open or multi-choice interview question - Play game - Rate game experience	- Evacuation of vulnerable people		- Risk perception	- Difference between risk perception of experts, lay persons and fire victims
Almeida et al. (2017)	- 3D - Interactive - Immersive - Game	16 + 62	- Room with corridors	Crowd	- Answer interview question	- Association between game and questionnaire - Evacuation of people with mobility impairment	- Sign - Smoke - Fire - Peer influence - Impairment type	- Direction choice	- How one can accommodate both heterogeneity and spatial correlation in exit choice
Butler et al. (2017)	- Interview	51	- Highrise buildings	N.A.	- Make choice in response to hypothetical emergency	- Cooperative behaviour - Proactive behaviour - Route choice	- Response type	- Likert scale	- Heterogeneity of passenger response - Passengers likelier to be reactive (than proactive) - Passengers likelier to be cooperative (than competitive) - Passengers unlikely to use escalators, lifts and tunnels - Tendency for herding behaviour not strong - Consumers reported more positive responses on all measured outcome variables after being exposed to the VR experience
Shiwakoti et al. (2017)	- Self-completed questionnaire	1134	- Urban train station	N.A.	- Rate satisfaction	- Crowding perception	- Relaxing treatment - Crowding level	- Perceived crowding - Mail satisfaction	- People would tend to choose evacuation lifts situation when suggested as the main exit option, and when stairs are overcrowded - When not guided, people prefer stairs - Being in the plaza (compared to the Mosque) predicted the extent of cooperation, but crowd density did not - Identifying the role of place and space in modulating processes of cooperation in crowds
Van Kerrebroeck et al. (2017)	- 3D - Immersive - HM display	167	- Room with four exits	N.A.	- Make choice between lifts and stairs	- Evacuation of tall buildings	- No. of people on stair/lift lobby - Refugee floor position - N.A.	- Choice	- Nearly half of participants chose extreme strategies by collecting either all or none of the objects before evacuating
Aleksandrov et al. (2018)	- Non-Immersive - Online SP Survey	566	- Tall buildings	Crowd	- Respond to questions (perceptions and ranks)	- Shared identity and cooperation among Hajj pilgrims		- Perceptions and ranks	
Alnabulsi et al. (2018)	- Verbally administered survey	1194	- Grand Mosque in Mecca	N.A.	- Make choice between collecting objects and leaving	- Pre-movement delay	- Knowledge of building - Social influence - Gain vs loss	- Number of objects collected - Odds of successful escape	
Bode and Codling (2018)	- 2D - Non-Immersive - Game	1200	- Room with two exits	Crowd					

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Table 4 (continued)

Author(s) (Year)	Method	Sample size	Hypothesised setup	Visualised (simulated) agents	Subject's task	Investigated phenomena	Design factors	Measured variables	Main observations or application
Chen et al. (2018e)	- Paper-based questionnaire	173	- Classroom with two exits	Crowd	- Make route choice	- Children evacuation behaviour	- Position - Congestion - Guidance - Group behaviour - Backtracking - Size of stimulus group	- Choice	- Position, congestion, group behaviour, and backtracking behaviour have significant effects on children route choice
Jorjafki et al. (2018)	- 3D - Immersive - Interactive - HM display	70	- Indoor location	Crowd	- Walk through an indoor location	- Behavioural contagion	- Exit familiarity - Social influence	- Prob of looking up - Proportion of time looking up	- Probability of looking up is dependent on the size of the stimulus group - Behavioural contagion can be triggered in the virtual environment
Kinatader et al. (2018)	- 3D - Immersive - Interactive - HM display	40	- Room with two exits	Crowd	- Make exit choice	- Exit choice	- Exit familiarity - Social influence	- Choice	- Exit familiarity and neighbour behaviour influence evacuation behaviour - Social influence increases with the number of neighbours
Lovreglio et al. (2018)	- 3D - Interactive - Immersive - Game - Vibrating platform	170	- Hospital building	Crowd	- Take appropriate action	- Earthquake preparedness and training	- Prototype type (behavioural vs training)	- Representation - Realism - Game interactivity	- Proposing a prototype VR SG training tool to enhance earthquake preparedness in public buildings
Moussaid et al. (2018)	- 3D - Interactive - Immersive - Multi-user	36	- Room leading to two corridors	Crowd	- Evacuate	- Bottleneck flow - Direction choice	- Urgency (stress) level	- Social influence - Choice - Flow rate - Density	- Describing methodological developments in VR crowd experiments, particularly multi-user platforms
Rio et al. (2018)	- 3D - Immersive - Interactive - HM display	56	- Open space	Crowd	- Walk with virtual crowd	- Social interaction in crowd motion	- Speed - Heading - Eccentricity - Distance	- Lateral deviation	- Neighbour influence decreases with distance, but not lateral position - Local neighbourhood of interaction is circularly symmetric
Shiwakoti et al. (2018)	- Self-completed questionnaire	1134	- Urban train station	N.A.	- Make choice in response to hypothetical emergency - Make exit choice	- Cooperative behaviour - Proactive behaviour - Route choice - Fire evacuation	- No. of neighbour - Response type	- Likert scale	- Passengers who report that they are likely to do nothing and/or push others if necessary, also report that they are less likely to get out safely
Tucker et al. (2018)	- 3D - Immersive - Interactive - HM display	39	- Library	Crowd			- Hazard level - Information	- Anxiety (stress) level (heart rate & skin response) - Choice of egress route - Collisions - Evacuation time	- Information about the obstacle decreased total evacuation time, but not choice of exit - Virtual hazard led to increased anxiety - Hazard level affected exit choice behaviour - No significant effect of hazard on evacuation time
Van den Berg et al. (2018)	- Interactive - Game	400	- Concert arena	Crowd	- Make choice to leave or stay	- Herding in decision to evacuate	- Social influence	- Choice to leave or stay	- The more people someone sees leaving, the more inclined this person is to leave - Seeing people leave has more impact than seeing people stay

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Table 4 (continued)

Author(s) (Year)	Method	Sample size	Hypothesised setup	Visualised (simulated) agents	Subject's task	Investigated phenomena	Design factors	Measured variables	Main observations or application
Arias et al. (2019b)	- 3D - Immersive - Interactive - HM display	111	- The Future Circular Collider	Solo	- Navigate	- Wayfinding evacuation systems	- Type of wayfinding system installation - Spatial exploration mode - Fire emergency	- Compliance to the instructions - Travel distance - Travel time	- Combination of red flashing lights, dynamic signage and a robot equipped with green flashing lights yielded the highest compliance - Participants under the fire emergency spent more time finding their way to exit than those under the control condition - Participants who did prior active exploration travelled longer in the egress task
Cao et al. (2019a)	- 3D - Immersive - Interactive - HM display	64	- Museum	Solo	- Navigate	- Wayfinding during fire emergency			
Deleca and Gofas (2019)	- 2D agent-based model & questionnaire	30	- Classroom	Crowd	- Navigate	- Student evacuation	- Door location - Desk location - Number of pupils - Sign colour	- Wayfinding decision	- Demonstrating possibility of using agent-based models to increase pupil's awareness of evacuation processes - Results challenged the local exposure hypothesis and supported the semantic association hypothesis
Kinatader et al. (2019)	- 3D - Immersive - HM display	24	- Room with two exits	Solo	- Identify the exit	- Illuminated emergency exit signs		- Choice	- Participants predominantly walked toward green signs
Li et al. (2019a)	- 2D - Interactive	146	- Passage with obstacles	Crowd	- Make route choice	- Route choice behaviour	- Obstacles configuration	- Choice	- Discrepancy between anticipated and actual evacuation behaviour - Comparison between virtual and field experiment of route choice - Qualitatively similar results in both field and computer (virtual) settings
Lin et al. (2019)	- 3D - Immersive - Interactive - HM display	60	- Museum	Solo	- Treasure hunting - Egress	- Indoor wayfinding	- Stress level - Exposure	- Wayfinding performance - Physiological response	- Significant positive impact of repeated exposure on wayfinding performance - Significant negative impact of mental stress caused by the fire emergency on wayfinding performance
Ronchi et al. (2019)	- 3D - Immersive - Interactive - HM display - & - CAVE	96 + 55	- Tunnel	Solo	- Navigate to safety	- Comparing VR methods (HMD vs. CAVE)	- VR method - Emergency sign type	- Likert scale scores	- Consistency between the two experimental methods

and people with mobility impairment including self-evacuation without assistance, assistance using a cane, assistance from another occupant or firefighter, or assistance using a stair chair. They used an evacuation drill dataset of 45 residents evacuating a six-story assisted-living facility and contrasted their speed measurements (0.11–0.29 m/s) with previous measurements in the literature. More on the evacuation characteristics of mobility-impaired individuals, Jiang et al. (2012) conducted an evacuation drill experiments with a sample of mixed-ability pedestrians and measured the unimpeded free walking speed of the participants during horizontal and stair ascending/descending movement. Lam et al. (2014) conducted an experimental study on upward movement in a high-rise building and observed that ascending speed was related to the age of the participants and that the older participants had relatively greater differences in blood pressure and heart rate before and after the experiment.

7.2. School/pre-school evacuation drills

In line with the topic of evacuation of vulnerable population, a number of studies reported school evacuation drills. The study of Najmanová and Ronchi (2017) presents an experimental data-set describing movement and behaviours of pre-school children consisting of travel speeds on horizontal and vertical escape routes (stairs), pre-evacuation times, and total evacuation times. They suggested that the main variables impacting the movement and the self-rescue abilities of children are their age and the configuration of escape route. In the study of Cuesta and Gwynne (2016), data of pre-evacuation times, travel speed and route use were collected during five evacuation drills from a school. Hamilton et al. (2017) focused, in their study, on the evacuation of primary schools. They investigated pre-evacuation times, movement on stairs and movement on the horizontal plane. They analysed data from twelve full scale evacuations of four primary schools (aged 4–12). They observed longer pre-evacuation times for classes of younger children and a reduction of the total evacuation time as a result of the repetition of the evacuation drill. Younger children were also observed to be slower on stairs in their observations. Studies of Cuesta et al. (2017), Poulos et al. (2018) and Chen et al. (2019) use school evacuation drill data for validation of agent-based models.

7.3. Evacuation drills in reduced visibility conditions

A number of evacuation drill experiments were conducted under reduced visibility conditions to simulate the conditions of fire evacuation. Zhang et al. (2018d) and Chen et al. (2018c) focused on the effect to limited visibility on the walking speed on stairs. Zhang et al. (2018d) observed that the descent speed sharply decreased when the visibility reduced, and that the group speed converged to the speed of the slowest person in the group. Chen et al. (2018c) observed that the influence of reduced visibility was much more significant on female pedestrians than males. They examined four different levels of visibility and observed that with the decrease of the visibility, the evacuation time first increased and then remained constant. Two other evacuation drill studies under reduced visibility were conducted in an underground facility (Jeon et al., 2011), examining the effect on evacuation time, movement speed and wayfinding; and in a supermarket (Cao et al., 2018a), observing typical behaviours such as following behaviour, helping behaviour and finding the dependent during the evacuation.

7.4. Social groups in evacuation drills

On the sub-topic of group behaviour on stairs, Chen et al. (2017a) investigated the relationship between evacuation speed and the load on pedestrians in a series of individual and group evacuation experiments under varying load conditions. They observed that the upward evacuation speed decreased as the floor increased while the downward evacuation speed changed very little with the floor. In group

evacuation, the average group evacuation speed decreased as the pedestrian load increased. Ma et al. (2017) focused mainly on the behaviour of small groups in tall building evacuations and observed that small groups can make positive or negative impact on crowd dynamics depending on the competitive mechanism.

7.5. Evacuation drill experiments of pedestrian ascent and descent on stairs

In other vertical evacuation drill studies, Chen et al. (2018a) investigated pedestrian ascent and descent under high occupancy condition and estimated movement speeds and flow rates. Chen et al. (2018b) validated a Pedestrian Footstep Model that they proposed to simulate pedestrian movement on stairways. Chen et al. (2017b) developed pedestrian ascent and descent fundamental diagram on stairway. Choi et al. (2014) studied how fatigue affects ascent and descent speed of pedestrians and observed that 50% of the population displayed a decrease in speed over the final half of the descent. However, nearly 50% of the population increased their travel speed during the final half of the descent. During the ascent, all participants decreased their speed over the first 20 floors by an average of approximately 60%. Lei et al. (2012a) compared simulated outcomes of FDS + Evac on stairs with experimental observations and found good agreement between the two. Huo et al. (2016) observed from the analysis of the fundamental diagram that the merging flow influences pedestrians' movement down stairs.

7.6. Evacuation drills in tunnels

Five studies were identified to have focused tunnel evacuation drills. Ronchi et al. (2018) conducted a tunnel evacuation experiment on movement speed and exit choice in smoke. They measured the walking speed in a certain range of non-irritant smoke condition. They also found a weak relationship between an individual's ability to walk in smoke and the unobstructed walking speed. The study of Seike et al. (2017) experimentally investigated the evacuation speed in a full-scale tunnel filled with smoke and clarified the relation between the extinction coefficient and normal walking speed. Porzycki et al. (2018) also performed a road tunnel evacuation drill in smoke condition and suggested that movement speed in smoke is influenced by the evacuees' attitude and familiarity with environment and evacuation procedures and not only by the visibility level. In other similar studies in this category, Zhang et al. (2018c) performed their drill experiments in vertical exit passages in an underwater road shield tunnel, and Zhang et al. (2019) conducted their evacuation drills using an extra-long steep-slope evacuation path in a high altitude tunnel fire.

7.7. Evacuation drills in large theatres and offices

Two recent studies reported on evacuation drills in large theatres. Galea et al. (2017a) analysed evacuation response behaviour of occupants during an announced evacuation drill in a large theatre during a live performance. Their results showed that the occupant response time distribution is related to the geometrical positioning of the occupants relative to proximity to exit aisles and exits. Response time is found to increase relative to the seat distance from the exit aisles and the distance of the seat row to an exit. The assumption is to large degrees in agreement with the foundation of the pre-evacuation time modelling method reported in the numerical study of Haghani et al. (2019c). Based on the analysis of walking trajectory datasets, Imanishi and Sano (2018) investigated route choice and flow rate in theatre evacuation drills. Their results showed that while the evacuees in the centre blocks chose the closest aisle from their seats, most of the evacuees in the side blocks chose the wall side aisles. More recently, Gao et al. (2019) conducted office evacuation drill experiments and observed abnormal individual behaviours that increased evacuation time. They also concluded that the personal response time plays a critical role in

evacuation process and in determining the evacuation time.

7.8. Evacuation drill experiments on signage systems

Two studies in this category address the topic of emergency signage. [Fu et al. \(2019\)](#) investigated how participants alone or in groups detect and accept the information conveyed by a signage system. The results of their study suggested that both signage detection and acceptance probabilities under individual conditions are larger than those under group situations. [Galea et al. \(2017b\)](#) evaluated the effectiveness of an improved active dynamic signage system based on their evacuation trials in a rail station. Their findings suggested that the improved system with voice alarm messaging that does not contradict the signage system successfully redirected 66% of the participants to the intended exit. The details of the evacuation drill studies identified in this review are provided and summarised in [Table 5](#).

8. Top conventional and emerging topics in pedestrian dynamics research

This updated review of the empirical crowd dynamics literature identified topics that, in line with the previous review, have been conventionally of great interest and been given priority by the researchers in this field. This determination is solely made based on the number of studies published on these topics. The review also identified topics that are gaining unprecedented attention and much further momentum compared to the previous review period. Although these topics may have been conventionally studied too, the amount of attention (reflected in the number of publications) paid to them has been substantially surged during the recent period of review in contrast to the previous review period. These two groups of research topics are respectively labelled in this work as “top conventional topics” and “top emerging topics”, and five items are identified within each category.

The top five conventional topics of crowd dynamics identified in this review are:

- (I) Bottleneck flow (faster is slower, obstacle at exit, exit location, exit capacity etc).
- (II) Walking behaviour (step taking, single-file movement, fundamental diagram, walking speed)
- (III) Conflicting flows (bidirectional flows, merging flows, intersecting flows, collision avoidance)
- (IV) Exit/route choice and navigation
- (V) Calibration and validation of models

The top five emerging topics of crowd dynamics according to this current review period are:

- (I) Evacuation of vulnerable people (mobility impairment and disability, elderly and children)
- (II) Evacuation training, education and guidance
- (III) Vertical evacuation (movement on stairs, evacuation of high-rise buildings)
- (IV) Group behaviour
- (V) Evacuation under limited visibility

These topics have also been summarised and illustrated in [Fig. 2](#). Please note that the numbering is simply a label and does not imply any particular ordering within each list. [Table 6](#) lists all the non-review studies that were surveyed in the current study and determines which of the top conventional or emerging topics have been addressed by each study. The studies have been categorised based on the type of experimental and data collection method. The readers may note that the conventional or emerging topics are not mutually exclusive. It is possible that a particular study has multiple dimensions and, therefore, has addressed multiple items in the list of the top topics. [Table 6](#)

data is in full recognition of that possibility as often publications had presented multiple dimensions and therefore were related to multiple categories in the list of top topics. The table in and of itself reveals interesting patterns as to the relation between various data collection methods and various top topics. These patterns will be looked into and further captured in more details in the section where the descriptive statistics are presented in Part II of this article.

9. Summary and conclusions

The extent of empirical research in the field of crowd dynamics has increased so substantially that has made it rather difficult for the researchers to stay abreast of this rapidly-evolving literature. Therefore, review and meta-analyses studies, whether narrowly focused or a covering broad range, might prove helpful in this area. Of course, at current time, reviews are more common than meta-analyses in this field. With few exceptions ([Bosina and Weidmann, 2017](#); [Lovreglio et al., 2019](#)), there have not been many classic meta-analyses published in this field. This could be partly attributable to the fact that, even though many topics are studied by various researchers, their design, execution and measurement methods ([Feliciani and Nishinari, 2018](#); [Li et al., 2019, 2020b](#)) are not invariably consistent enough to make a meta-analysis possible. Some areas of research where meta-analyses and meta-syntheses could potentially be of great value include measurements of exit/corridor capacities (under various influential factors such as width, crowd conduct etc), or pedestrian speed (under various conditions and for various populations) or evacuation/pre-evacuation time ([Zhao et al., 2020a](#)). These measurements are essential and fundamental to the finetuning of numerical methods. With the accumulation of the empirical the measurements, it seems to be timely and warranted to aggregate these measurements in the form of meta-analysis/synthesis studies. The scarcity of adequate and informative meta-analyses could in fact be one of the major knowledge gaps in the current literature of crowd dynamics. It could be partly due to the inconsistencies in the measurement/experimentation methods or the scatters of the topics in this field. The current review, however, suggests that the experimental and measurement methods are becoming increasingly harmonious, focusing on clusters of readily-identifiable research themes. This may pave the way for informative meta-analyses in the near future.

The analyses of the new empirical studies in this work, in contrast with the previous survey of the literature further revealed that some data collection methods are becoming more popular and some are used relatively less frequently compared to the previous years. In particular, the share of the studies based on animal/insect crowd experiments in the empirical literature appears to have considerably shrunk compared to the previous review period. Instead, more focus has been placed on laboratory experiments with crowds of humans, a method that nearly constituted half of all studies surveyed in this updated review. The methods of evacuation drill and VR/AR experiments had maintained their standing in the empirical literature and in addition, had the played the most instrumental role, among all seven empirical methods, in addressing and exploring what was identified in this work as top emerging topics in the crowd literature. Laboratory experiments with human crowds, on the other hand, appear to have played a significantly bigger role in investigating the top conventional topics.

The survey of the recent studies made it also clear that the literature has started to pay unprecedented attention to a number of new topics, those that did not used to be heavily studied and of which only scattered publications had been identified in the previous review. Growing attention is going to wards gaining empirical insight into the characteristics of the evacuation of vulnerable population (e.g. children, elderly and people with mobility impairment ([Feliciani et al., 2020](#); [Georg et al., 2019](#))). More studies are looking into the potential role and benefits of evacuation training, education and guidance, a topic that was not in the focus of attention until the recent years. Evacuation of high-rise buildings and behavioural aspects specific to vertical

Table 5
Studies based on evacuation drill experiments in chronological order.

Author(s) (Year)	No. of evacuees	Annou.?	Venue type	General vision	Crowd. level	Investigated phenomena	Measured variables	Post-drill interview?	Main finding or application
Gwynne et al. (2005)	100 & 381	Yes	Small room in a school gym + Milbourn House (eight story building)	Intact	High	Evacuation movement	- Total evacuation time - Exit/stair choice - Response time	Yes	- Validation of the buildingEXODUS evacuation model
Jeon et al. (2011)	Up to 32	Yes	Underground facilities	Intact and limited	High	Evacuation under limited visibility	- Evacuation time - Movement speed - Wayfinding - Evacuation routes	Yes	- Difference in visibility condition caused changes in evacuee's movement speed and travel distance. - Change in visibility condition by indoor ordinary lights caused significant change only in movement speed on flat floors - Change in visibility by smoke caused significant change in travel distance as well as movement speed - Obtaining basic data of free movement speed of the mobility impaired pedestrians on horizontal ground & stair
Jiang et al. (2012)	100	Yes	Subway station	Intact	Low	Walking speed of people with mobility impairment on stairs	- Speed	Yes	- Simulation model validation
Lei et al. (2012a)	384	Yes	Dormitory	Intact	High	Exit and stair condition	- Flow rate	No	- Simulation model validation
Lei et al. (2012b)	240	Yes	Dormitory	Intact	High	Pre-movement time modelling	- Evacuation time - Evacuation times	No	- Investigating the effect of including a pre-movement time module on the simulation accuracy
Kuligowski et al. (2013)	40	Yes	Classroom	Intact	High	Stair evacuation of elderly and mobility impaired	- Evacuation time - Exit usage	No	- Investigating the effect of occupant density around exits on the behaviour
Choi et al. (2014)	60	Yes	50-story building	Intact	Solo	Walking on stairs	- Speed	No	- Simulation validation
Lam et al. (2014)	120	Yes	High-rise building	Intact	Solo	Upward movement on stairs	- Speed - Age - Blood pressure	No	- Average descent speed for the male and female population was 0.83 m/s and 0.74 m/s - Average ascent speed was 0.66 m/s and 0.48 m/s for male and female respectively
Cuesta and Gwynne (2016)	300	No	School building	Intact	High	School evacuation	- Pre-evacuation time - Travel speed - Route use	No	- Ascending speed was related to the age of the participants - Identifying the typical and unique movement characteristic of Hong Kong people
Huo et al. (2016)	73	Yes	High-rise building	Intact	High	Walking on stairs	- Heartbeat rate - Travel speed	No	- Data on vulnerable population's (school student) evacuation
Mu et al. (2016)	370	Yes	Building	Intact	High	Imbalance in exit choice	- Speed - Fundamental diagram	No	- Analysing microscopic movement characteristics of pedestrians on stair landings
Zhu et al. (2016)	98	Yes	Classroom	Intact	High	Evacuation from rooms with obstacle	- Flow - Density	No	- Merging flow influences pedestrians' movement down stairs - The two doors were not used evenly
Chen et al. (2017a)	120	Yes	26-story residential building	Intact	Solo and High	Walking on stairs(without a bag with varying load)	- Evacuation time - Speed	No	- Calibrating social force model - Comparing simulation and experimental data
Chen et al. (2017b)	30	Yes	Staircase	Intact	Solo to high	Fundamental diagram on stairway	- Speed - Density	No	- Upward evacuation speed decreased as the floor increased while the downward evacuation speed changed very little
Cuesta et al. (2017)	300	No	School building	Intact	High	School evacuation	- Pre-evacuation time - Travel speed	No	- Average group evacuation speed decreased as the pedestrian load increased
Galea et al. (2017a)	1200	No	Large theatre	Intact	High	Response behaviour	- Route use - Response time	No	- Analysing stair movement of pedestrians carrying a bag - with the increase of pedestrian density, the speed decrease rate varies - Validating various egress modelling tools

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Table 5 (continued)

Author(s) (Year)	No. of evacuees	Annou.?	Venue type	General vision	Crowd. level	Investigated phenomena	Measured variables	Post-drill interview?	Main finding or application
Galea et al. (2017b)	Up to 170	Yes	Rail station	Intact	High	Intelligent active dynamic signage system	- Exit usage	No	- Improved ADSS with voice alarm messaging that does not contradict the signage system successfully redirected 66% of the participants to intended exit
Hamilton et al. (2017)	Up to 250	Yes	School buildings	Intact	High	School evacuation (pre-evacuation time, stairs, plane)	- Pre-evacuation time - Speed	No	- Longer pre-evacuation times obtained younger children - A reduction in pre-evacuation times was observed through repetition of evacuation drills - On stairways children in the lower classes moved slower and were more cautious than older children
Ma et al. (2017)	30 & 56	Yes	11-story office building	Intact	High	Small groups	- Speed - Density	No	- Small groups make positive or negative impacts on crowd dynamics depending on the competitive mechanism
Najmanová and Ronchi (2017)	188	No	School building	Intact	High	Pre-school evacuation (stairs, plane)	- Speed - Pre-evacuation time - Total evacuation time	No	- Main variables impacting the movement and the self-rescue abilities of children are age and the configuration of the escape route
Seike et al. (2017)	120	Yes	Tunnel	Dark + Smoke	High	Evacuation speed in tunnel evacuation	- Speed	No	- Investigating the relation between extinction coefficient and normal walking speed
Zeng et al. (2017)	39	Yes	9-story building	Dark and intact	Solo	Movement on stairs	- Stair preference - Speed	No	- Normal and evacuation speed influenced by darkened space - Studying the velocities and preference with different illumination
Cao et al. (2018a)	65	Yes	Supermarket	Intact and blinded	High	Evacuation under limited visibility	- Velocity - Exit choice	Yes	- Preference for stairs will change with illuminations - Following behaviour, helping behaviour and finding the dependent during evacuation under limited visibility are observed from - Reduced visibility hinders pedestrian movement and prolongs evacuation process
Chen et al. (2018a)	Up to 45	Yes	20-level building	Intact	High	Movement on stairs	- Flow - Speed - Density	No	- Studying ascent and descent movement of people on long stair with high occupant density
Chen et al. (2018b)	20	Yes	Staircase	Intact	High	Single-file movement on stairs	- Speed - Density	No	- Validating and calibrating a numerical simulation method
Chen et al. (2018c)	120	Yes	26-floor residential building	Intact & reduced	Solo & high	Movement on stairs	- Speed - Density	No	- The influence of visibility was more significant on females - With the decrease of visibility, evacuation time increased first and then levelled out
Hu et al. (2018)	126	Yes	School building	Intact	High	Classroom evacuation	- Exit choice	No	- Validating a numerical simulation method - At low pedestrian density, evacuation efficiency can be improved through adoption of the shortest route strategy
Imanishi and Sano (2018)	540	Yes	Theatre	Intact	High	Route choice	- Route choice	No	- Collecting data for validation of simulation models - While the evacuees in the centre blocks chose the closest aisle from their seats, most of the evacuees in the side blocks chose the wall side aisles - Specific flow at exits was 0.96 per/m/s in congested situation

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Table 5 (continued)

Author(s) (Year)	No. of evacuees	Annou.?	Venue type	General vision	Crowd. level	Investigated phenomena	Measured variables	Post-drill interview?	Main finding or application
Porzycki et al. (2018)	28 & 50	Yes	Tunnel	Smoke & intact	High	Tunnel evacuation under reduced visibility	- Evacuation time - Pre-movement time - Speed	No	- Shorter pre-movement time and shorter evacuation time were observed as a result of learning - Movement speed in smoke is influenced by familiarity with environment in addition to the visibility level
Poulos et al. (2018)	1500	Yes	School	Intact	High	School evacuation	- Flow rate	No	- Validation of an agent-based building evacuation model
Ronchi et al. (2018)	66	Yes	Tunnel	Smoke	Solo	Tunnel evacuation under limited visibility	- Speed - Exit choice	No	- There is a weak relationship between an individual's ability to walk in smoke and the unobstructed walking speed
Zhang et al. (2018d)	75	Yes	Five-story office building	Intact & reduced	Solo and groups	Movement on stairs under limited visibility	- Speed	No	- Study the stairwell evacuation performance of individuals and small groups under normal and impaired visibility conditions
Zhang et al. (2018c)	48	Yes	Underwater road tunnel	Intact	High	Tunnel evacuation and movement on stairs	- Exit time - Speed	No	- The descent speed decreased sharply when the visibility was reduced - Group behaviour had negative effect on evacuation during normal visibility - The group behaviour resulted in quicker movement speed in low visibility condition - Individual movement speed in staircase was lower than the speed in slide - Average Occupant Flow Rate (OFR) in staircase was higher than the OFR in slide - The upstairs movement of firefighters did not negatively affect evacuation of occupants down stairs - Training or drills can help improve the movement speed of occupants
Chen et al. (2019)	52	Yes	Classroom	Intact	High	School evacuation	- Evacuation time - Exit choice	No	- Validation of an agent-based building evacuation model - Identifying several typical patterns of child behaviour related to distance, obstacle and congestion level
Fu et al. (2019)	119	Yes	University building	Intact	High	Emergency signage	- Route choice - Decision time	Yes	- Both signage detection and acceptance probabilities under individual conditions are larger than those with groups - High-placed sign has positive effect on route choice
Zhang et al. (2019)	20	Yes	Steep-slope Tunnel	Intact	High	Tunnel evacuation	- Response time - Route choice - Movement time - Speed change	Yes	- Investigating pedestrian evacuation using an extra-long steep-slope evacuation path in a high-altitude tunnel fire

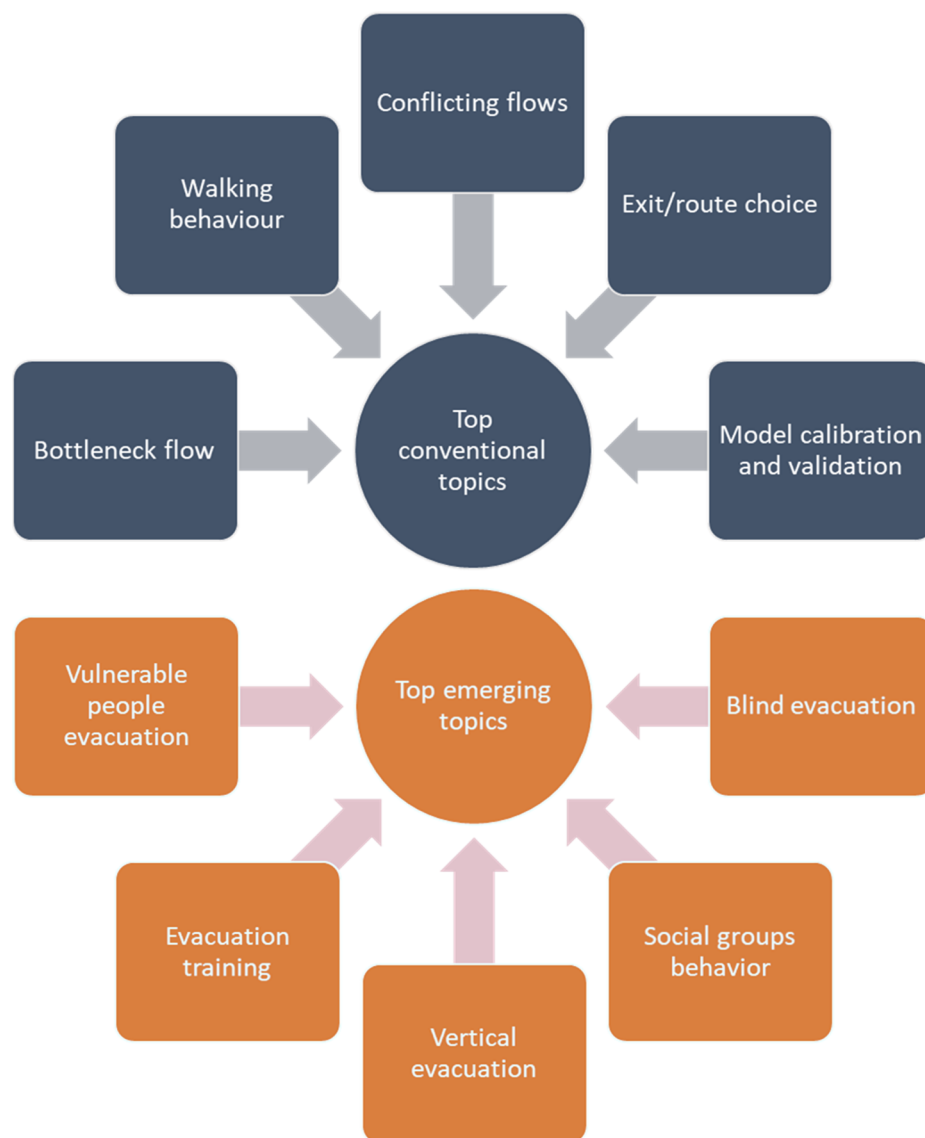


Fig. 2. Top conventional and top emerging topics in the contemporary literature of empirical crowd dynamics research.

evacuations (e.g. movement on stairs, the choice between emergency stairs and refuge floors and elevators) has also constituted the main topic for a considerable number of journal articles and even doctorate dissertations (Aleksandrov, 2018). Questions around the topic of social groups in crowd dynamics (Do et al., 2016) and evacuation under reduced visibility condition both did use to be conventionally studied but gained distinct growing attention in the new review period. These topics were identified as top five emerging topics in this work. Although many more emerging topics could as well be identified, these five items appeared to have been dominant and more noticeable solely based on the number of publications.

It was not an intention of this review to list a suggestion of potential topics for future research as these topics need to emerge naturally based on the judgment of individual researchers. Plus, attempting to make such a list in a broad context may result in an awfully long and possibly arbitrary list of potential topics. However, just on a closing note, the author would just like to highlight one potential topic that has started to gain momentum but needs further attention and this is the issue of external validity, replicability and transferability of the experimental findings across methods, across populations, across physical layouts, and across other contextual factors. Evidence on these topics do currently exist but are rather sparse (Cao et al., 2018b; Haghani and Sarvi,

2019c; Jin et al., 2019a; Li et al., 2019a; Moussaïd et al., 2016) and the topic seems to be deserving much further attention.

Given that researchers in this field use a variety of methods for experimentation, it is important that we know whether or when these findings are replicable across methods. This is particularly important for when we face multiple avenues of experimentation (e.g. laboratory crowd vs VR/AR) where we need to make a determination and trade-off between the logistical and monetary costs of experimentation, their level of realism, and their level of experimental control. Furthermore, VR or laboratory crowd experiments are performed in certain layouts (e.g. certain building configurations) with a certain group of participants. It is of great importance to know to what extent the outcomes are only limited to that particular physical layout or type of population/demographic (Haghani and Sarvi, 2019c; Subaih et al., 2020; Zhu et al., 2020).

While it is important to establish reliable borders of applicability and generalisability for these experimental findings and to avoid over-generalisation, it is also equally important to not cast unnecessary amounts of doubts over their generalisability. For example, unless there are solid reasons and reasonable justifications that nationality, age or gender or the level of physical fitness play a significant role in a particular experimental task, there is no need to be excessively cautious

Table 6

The list of experimental studies of crowd dynamics surveyed in this current review along with their relevance to the top conventional or top emerging topics. Studies within each category of data collection are listed in chronological order.

No.	Data collection type	Reference	Top five conventional topics					Top five emerging topics				
			I	II	III	IV	V	I	II	III	IV	V
1	Anim/Ins	Shiwakoti et al. (2014)	•									
2	Anim/Ins	Chung and Lin (2017)				•						
3	Anim/Ins	Lin et al. (2017)	•									
4	Anim/Ins	Oh and Park (2017)	•									
5	Anim/Ins	Shahhoseini and Sarvi (2017)			•							
6	Anim/Ins	Zhang et al. (2017)	•									
7	Anim/Ins	Chen et al. (2018d)	•									
8	Anim/Ins	Ji et al. (2018b)				•						
9	Anim/Ins	Wang et al. (2018a)	•									
10	Anim/Ins	Wang et al. (2018d)			•							
11	Anim/Ins	Wu et al. (2018)	•									
12	Anim/Ins	Zhang et al. (2018b)	•									
13	Anim/Ins	Wang et al. (2019c)		•								
14	Anim/Ins	Xiao et al. (2019a)		•								
15	Laboratory	Isobe et al. (2004)				•						•
16	Laboratory	Hoogendoorn and Daamen (2005)	•									
17	Laboratory	Cepolina (2009)	•									
18	Laboratory	Yanagisawa et al. (2009)	•									
19	Laboratory	Ma et al. (2010)		•								
20	Laboratory	Yanagisawa et al. (2010)	•									
21	Laboratory	Köster et al. (2011)									•	
22	Laboratory	Jelić et al. (2012)		•								
23	Laboratory	Yanagisawa et al. (2012)		•								
24	Laboratory	Burghardt et al. (2013)								•		
25	Laboratory	Guo et al. (2013)				•						
26	Laboratory	Hrabák et al. (2013)	•									
27	Laboratory	Jiang et al. (2014)	•									
28	Laboratory	Shimura et al. (2014)						•				
29	Laboratory	Wolinski et al. (2014)			•		•					
30	Laboratory	Cao et al. (2015)										•
31	Laboratory	Fernández et al. (2015)	•									
32	Laboratory	Sharifi et al. (2015)		•				•		•		
33	Laboratory	Werberich et al. (2015)				•	•					
34	Laboratory	Cao et al. (2016)		•				•				
35	Laboratory	Crociani et al. (2016)				•	•					
36	Laboratory	Cuesta et al. (2016)										
37	Laboratory	Ezaki et al. (2016)										
38	Laboratory	Feliciani and Nishinari (2016)			•							
39	Laboratory	Garcimartín et al. (2016)	•									
40	Laboratory	Gorrini et al. (2016)			•						•	
41	Laboratory	Guo et al. (2016)			•		•					•
42	Laboratory	Liao et al. (2016)	•									
43	Laboratory	Nikolić et al. (2016)	•									
44	Laboratory	Parisi et al. (2016)			•							
45	Laboratory	Sharifi et al. (2016)						•				
46	Laboratory	Colombi and Scianna (2017)				•						
47	Laboratory	Garcimartín et al. (2017)	•									
48	Laboratory	Guo et al. (2017)	•									
49	Laboratory	Hänseler et al. (2017)			•							
50	Laboratory	Lian et al. (2017)			•							
51	Laboratory	Liao et al. (2017b)	•				•					
52	Laboratory	Liu et al. (2017)					•					
53	Laboratory	Lu et al. (2017)					•				•	
54	Laboratory	Rahman et al. (2017)		•								
55	Laboratory	Roy et al. (2017)			•							
56	Laboratory	Seriani et al. (2017)	•									
57	Laboratory	Shahhoseini et al. (2017)			•							
58	Laboratory	Sharifi et al. (2017)		•				•				
59	Laboratory	Sieben et al. (2017)	•									
60	Laboratory	Xue et al. (2017)			•							
61	Laboratory	Yuan et al. (2017)			•							
62	Laboratory	Zhao et al. (2017)		•								
63	Laboratory	Zhao and Zhang (2017)		•								
64	Laboratory	Adrian et al. (2018)	•									
65	Laboratory	Bukáček et al. (2018)	•									
66	Laboratory	Cao et al. (2018b)			•							
67	Laboratory	Cao et al. (2018c)		•								
68	Laboratory	Crociani et al. (2018)									•	
69	Laboratory	Dias and Lovreglio (2018)		•								
70	Laboratory	Gaire et al. (2018)				•		•				
71	Laboratory	Garcimartín et al. (2018)	•									

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Table 6 (continued)

No.	Data collection type	Reference	Top five conventional topics					Top five emerging topics				
			I	II	III	IV	V	I	II	III	IV	V
72	Laboratory	Gwynne et al. (2018)										
73	Laboratory	Haghani and Sarvi (2018b)				•						
74	Laboratory	Huang et al. (2018a)		•								
75	Laboratory	Huang et al. (2018b)		•								
76	Laboratory	Ji et al. (2018a)										
77	Laboratory	Jin et al. (2018)		•								
78	Laboratory	Kim et al. (2018)		•								
79	Laboratory	Lian et al. (2018a)										
80	Laboratory	Lian et al. (2018b)										
81	Laboratory	Liu et al. (2018)					•					
82	Laboratory	Ma et al. (2018)		•								
83	Laboratory	Nagao et al. (2018)										
84	Laboratory	Seriani and Fujiyama (2018)	•									
85	Laboratory	Shahhoseini and Sarvi (2018)			•							
86	Laboratory	Shahhoseini et al. (2018)	•		•							
87	Laboratory	Song et al. (2018a)		•			•					
88	Laboratory	Song et al. (2018b)					•					
89	Laboratory	Sun et al. (2018)		•								
90	Laboratory	Templeton et al. (2018)		•								
91	Laboratory	Wang and Weng (2018)										
92	Laboratory	Wang et al. (2018b)		•								
93	Laboratory	Wang et al. (2018c)		•								
94	Laboratory	Zeng et al. (2018)		•								
95	Laboratory	Zhang et al. (2018a)		•								
96	Laboratory	Bode et al. (2019)			•							
97	Laboratory	Cao et al. (2019b)										•
98	Laboratory	Cao et al. (2019c)		•								
99	Laboratory	Georg et al. (2019)	•					•				
100	Laboratory	Haghani and Sarvi (2019b)				•						
101	Laboratory	Haghani and Sarvi (2019a)				•						
102	Laboratory	Haghani and Sarvi (2019e)				•						
103	Laboratory	Haghani and Sarvi (2019d)				•						
104	Laboratory	Haghani et al. (2019d)	•									
105	Laboratory	Haghani et al. (2019e)				•					•	
106	Laboratory	Hu et al. (2019)			•							
107	Laboratory	Huang et al. (2019)		•								
108	Laboratory	Jia et al. (2019)		•								
109	Laboratory	Jin et al. (2019a)		•								
110	Laboratory	Nicolas et al. (2019)										
111	Laboratory	Köster et al. (2019)								•		
112	Laboratory	Seriani et al. (2019)	•									
113	Laboratory	Shahhoseini and Sarvi (2019)			•							
114	Laboratory	Shi et al. (2019)	•									
115	Laboratory	Stuart et al. (2019)		•				•				
116	Laboratory	Tavana and Aghabayk (2019)	•									
117	Laboratory	Wang et al. (2019a)		•								
118	Laboratory	Wang et al. (2019b)			•							
119	Laboratory	Xiao et al. (2019b)			•							
120	Laboratory	Yamamoto et al. (2019)			•							
121	Laboratory	Zeng et al. (2019)		•								
122	VR/AR/Hyp	Aoki et al. (2007)				•			•			
123	VR/AR/Hyp	Smith and Ericson (2009)						•	•			
124	VR/AR/Hyp	Kinateder et al. (2013)							•			
125	VR/AR/Hyp	Duarte et al. (2014)							•			
126	VR/AR/Hyp	Meng and Zhang (2014)				•						
127	VR/AR/Hyp	Chittaro and Sioni (2015)							•			
128	VR/AR/Hyp	Gamberini et al. (2015)										
129	VR/AR/Hyp	Lovreglio et al. (2015)					•					
130	VR/AR/Hyp	Burigat and Chittaro (2016)				•			•			
131	VR/AR/Hyp	Cosma et al. (2016)				•						•
132	VR/AR/Hyp	Crescentini et al. (2016)							•			
133	VR/AR/Hyp	Kinateder and Warren (2016)										
134	VR/AR/Hyp	Lovreglio et al. (2016a)				•						
135	VR/AR/Hyp	Tancogne-Dejean and Laclémence (2016)						•				
136	VR/AR/Hyp	Almeida et al. (2017)				•						
137	VR/AR/Hyp	Butler et al. (2017)						•				
138	VR/AR/Hyp	Shiwakoti et al. (2017)				•						
139	VR/AR/Hyp	Van Kerrebroeck et al. (2017)										
140	VR/AR/Hyp	Aleksandrov et al. (2018)								•		
141	VR/AR/Hyp	Alnabulsi et al. (2018)										
142	VR/AR/Hyp	Bode and Codling (2018)										
143	VR/AR/Hyp	Chen et al. (2018e)				•						
144	VR/AR/Hyp	Jorjafki et al. (2018)										

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Table 6 (continued)

No.	Data collection type	Reference	Top five conventional topics					Top five emerging topics				
			I	II	III	IV	V	I	II	III	IV	V
145	VR/AR/Hyp	Kinateder et al. (2018)				•						
146	VR/AR/Hyp	Lovreglio et al. (2018)							•			
147	VR/AR/Hyp	Moussaïd et al. (2018)	•			•						
148	VR/AR/Hyp	Rio et al. (2018)		•								
149	VR/AR/Hyp	Shiwakoti et al. (2018)				•						
150	VR/AR/Hyp	Tucker et al. (2018)				•						
151	VR/AR/Hyp	Van den Berg et al. (2018)										
152	VR/AR/Hyp	Arias et al. (2019b)				•						
153	VR/AR/Hyp	Cao et al. (2019a)				•						
154	VR/AR/Hyp	Delcea and Cotfas (2019)				•						
155	VR/AR/Hyp	Kinateder et al. (2019)										
156	VR/AR/Hyp	Li et al. (2019a)				•						
157	VR/AR/Hyp	Lin et al. (2019)				•			•			
158	VR/AR/Hyp	Ronchi et al. (2019)				•			•			
159	Evac Drills	Gwynne et al. (2005)					•					
160	Evac Drills	Jeon et al. (2011)										•
161	Evac Drills	Jiang et al. (2012)								•		•
162	Evac Drills	Lei et al. (2012a)				•				•		
163	Evac Drills	Lei et al. (2012b)					•					
164	Evac Drills	Kuligowski et al. (2013)						•		•		
165	Evac Drills	Choi et al. (2014)								•		
166	Evac Drills	Lam et al. (2014)								•		
167	Evac Drills	Cuesta and Gwynne (2016)					•	•				
168	Evac Drills	Huo et al. (2016)								•		
169	Evac Drills	Mu et al. (2016)				•	•					
170	Evac Drills	Zhu et al. (2016)					•					
171	Evac Drills	Chen et al. (2017a)								•	•	
172	Evac Drills	Chen et al. (2017b)								•		
173	Evac Drills	Cuesta et al. (2017)				•	•	•				
174	Evac Drills	Galea et al. (2017a)										
175	Evac Drills	Galea et al. (2017b)							•			
176	Evac Drills	Hamilton et al. (2017)						•		•		
177	Evac Drills	Ma et al. (2017)								•	•	
178	Evac Drills	Najmanová and Ronchi (2017)						•				
179	Evac Drills	Seike et al. (2017)										•
180	Evac Drills	Zeng et al. (2017)								•		•
181	Evac Drills	Cao et al. (2018a)										•
182	Evac Drills	Chen et al. (2018a)								•		
183	Evac Drills	Chen et al. (2018b)								•		
184	Evac Drills	Chen et al. (2018c)								•		•
185	Evac Drills	Hu et al. (2018)					•	•				
186	Evac Drills	Imanishi and Sano (2018)				•						
187	Evac Drills	Porzycki et al. (2018)										•
188	Evac Drills	Poulos et al. (2018)					•	•				
189	Evac Drills	Ronchi et al. (2018)		•		•						•
190	Evac Drills	Zhang et al. (2018d)								•	•	•
191	Evac Drills	Zhang et al. (2018c)								•		
192	Evac Drills	Chen et al. (2019)						•				
193	Evac Drills	Fu et al. (2019)							•			
194	Evac Drills	Zhang et al. (2019)								•		

about generalisation across demographics. After all, these experiments are only approximations of the behaviour and if experimental findings with a group of young fit students of certain nationality could reasonably approximate other demographics, then there is no reason to repeat those experiments across demographics. On the other hand, though, when the task is significantly correlated with the physical ability or fitness, then we may choose to collect a demographically representative sample or replicate the experiments across various demographics. Therefore, it is not suggested here that every single experiment has to be replicated by different methods, or different populations or in different layouts. But tests of external validity and transferability, when designed thoughtfully, could be of great value.

Thus far, the existing limited evidence has been very promising in favour of the generalisability of the experimental findings. But we also need to take these evidences with a certain degree of caution too and question if there could exist a publication bias effect. Clearly when testing the problem of external validity, the findings are much likelier to be publishable if they produce positive outcomes rather than

negative outcomes; and that could potentially affect our perception of the level of generalisability in a biased way, if the literature only keeps admitting studies that produce positive findings. In the absence of enough evidence, there is always the possibility of remaining overly optimistic or being excessively cautious about the level of generalisability, which are both undesirable. But, as mentioned, evidence on this aspect is gradually accumulating and with further attention, we may be able to establish a clearer level of confidence in experimental findings in the field of crowd dynamics.

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